Random Variables

Samatrix Consulting Pvt Ltd

Samatrix.io

Random Variables

- So far, we have studied the random experiments and events.
- We have also studied the concepts of probability for random events.
- We have seen how conditional probability can change our belief about an unobserved event, given that we have new observed new evidence or related event.
- In this section, we will study discrete and continuous random variables and probability distributions.
- For example, in the case of four tosses of a coin, the number of heads could be any one of the following possible values: $0, 1, 2, 3, 4$.
- The random variable, in this case, is the number of heads that may be one of the possible values, with a distribution of probabilities over this set of values.
- In other words, we can say that a random variable describes the outcome of the experiment in terms of a number.

Random Variables

- When conducting a random experiment, the experimenter takes a measurement.
- This measurement is the outcome of a random variable.
- We are interested in the probability of measurement that will result in A whereas A is the subset of outcome space of X.
- If we know the probability of measurement for all subsets A, we know the probability distribution of the random variable.
- Generally, we denote the random variables using capital letters X, Y, Z, etc.
- The value the random variable takes is denoted by lowercase letters, e.g., x,y, z, etc.
- For example, we can use X for “the number obtained by rolling a die”, Y for "the number of heads in four-coin tosses", and Z for "the suit of a card dealt from a well-shuffled deck".
- The range of a random variable X is the set of all the possible values that X might produce.

Random Variables and Their Range

Random Variable	Description	Range
X	Number on a Die	{1, 2, 3, 4, 5, 6}
Y	Number of heads in 4 coin toss	{0, 1, 2, 3, 4}

Discrete Random Variables

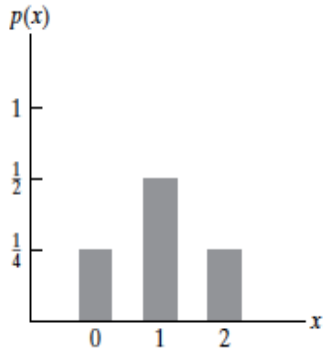
Discrete Random Variables

- The dictionary meaning of discrete is distinct or separate.
- Going by this meaning, the random variables Y that can take on distinct or separated values y_k are called discrete random variables.
- The possible number of values can be finite, for example, the number of heads in 5 tosses has possible values $0,1,2,3,4,5$.
- There can also be a countably infinite number of possible values for example number of tosses until the first head has possible values $0,1,2,\cdots,\infty$.
- In the case of discrete random variables, the possible values are separated by gaps.

Probability Mass Function

- For a discrete random variable, we can define the probability mass function
$$f(y_k) = P(Y = y_k)$$
- For the countable number of values of y_k, the probability mass function $f(y_k)$ is positive. For all possible values y_k of the discrete random variable,
$$f(y_k) \geq 0 \quad \text{for } k = 1, 2, \cdots$$
$$f(y) = 0 \text{ for all values of } y$$
- Since Y must take on one of the values y_k
$$\sum_{i=1}^{\infty} f(y_k) = 1$$
- We can present the probability mass function in a graphical format

Example



- Suppose we conduct an experiment by tossing 2 fair coins. If Y denotes the number of heads appear on the top. The random variable, Y, can take one of the values, $0, 1,$ and 2. The probabilities are

$$P\{Y = 0\} = P\{(T, T)\} = \frac{1}{4}$$

$$P\{Y = 1\} = P\{(T, H), (H, T)\} = \frac{1}{2}$$

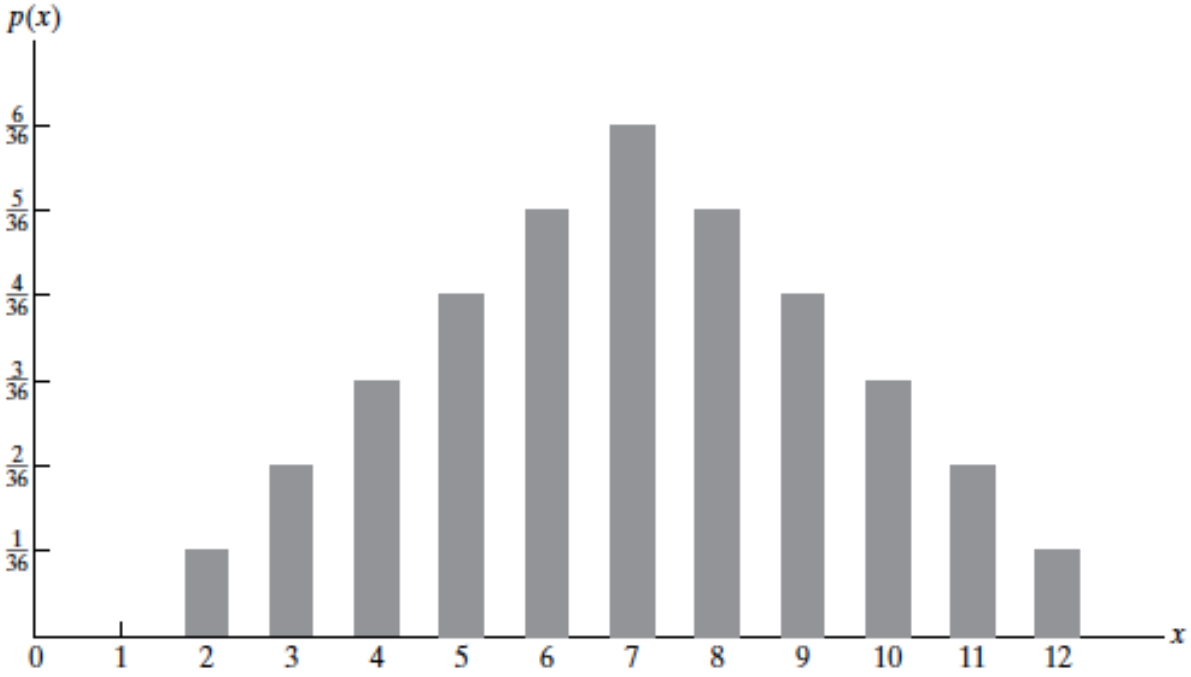
$$P\{Y = 2\} = P\{(H, H)\} = \frac{1}{4}$$

- We can present the probability mass function by plotting in a graphical format

Samatrix.io

Example

- Similarly, we can represent the probability mass function of a random variable representing the sum when we roll two dice as a part of our experiment

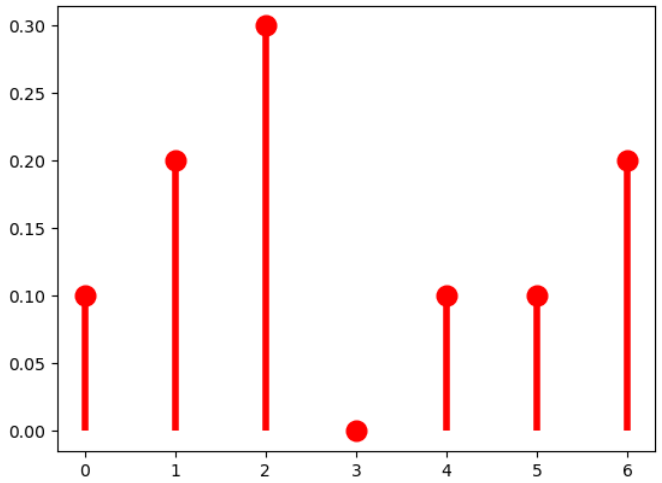


Samatrix.io

Discrete Distribution using Python

- We can use scipy.stats.rv_discrete()to construct discrete distribution by a list of values and corresponding probabilities.
-

```
>>> import numpy as np
>>> from scipy import stats
>>> import matplotlib.pyplot as plt
>>> yk = np.arange(7)
>>> pk = (0.1, 0.2, 0.3, 0.0, 0.1, 0.1, 0.2)
>>> custm = stats.rv_discrete(name='custm', values=(yk, pk))
>>> fig, ax = plt.subplots(1,1)
>>> ax.plot(yk, custm.pmf(yk),'ro',ms=12,mec='r')
[<matplotlib.lines.Line2D object at 0x7feb9d140fd0>]
>>> ax.vlines(yk,0,custm.pmf(yk),colors='r',lw=4)
<matplotlib.collections.LineCollection object at 0x7feb9d1551f0>
>>> plt.show()
```



Expectations

- We can define the expected value of a discrete random variable Y as the sum of each possible values times its probability.

$$E(Y) = \sum_{k=1} y_k \times f(y_k)$$

- We also call the expected value of the random variable Y as the mean of the random variable and denote by μ. We can denote the sample mean of a random sample of size n is

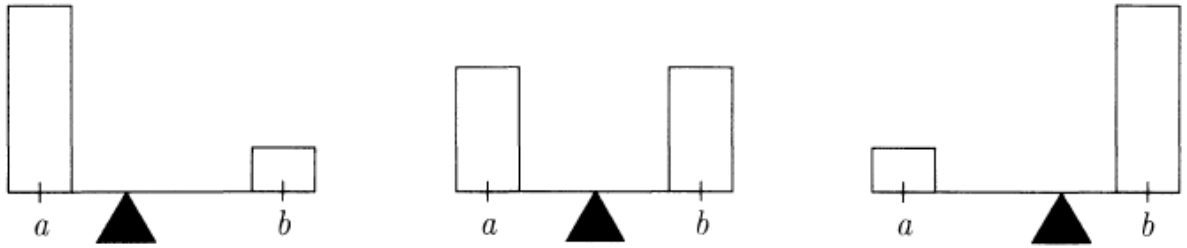
$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

We can derive the expected value from the distribution of Y the same way as we derive the mean or average $\bar{y}$ or μ of a list. For example, the average of the list $(1,0,8,6,6,1,6)$ of $n = 7$ number is

$$\frac{1+0+8+6+6+1+6}{7} = 0 \times \frac{1}{7} + 1 \times \frac{2}{7} + 6 \times \frac{3}{7} + 8 \times \frac{1}{7} = 4$$

Example

- If random variable Y can take two possible values: a and b, with the probabilities $P(a)$ and $P(b)$. Then
- $E(Y) = a\,P(a) + bP(b)$
- Where $P(a) + P(b) = 1$. The weighted average of a and b would be a number between a and b. The larger $P(a)$, the closer $E(Y)$ is to a; and the larger $P(b)$, the closer $E(Y)$ is to b



Samatrix.io

Variance

- If you predict the value of a random variable Y using its expected value $E(Y) = \mu$, you will be off by the random amount $Y - \mu$ which is known as deviation

$$E(Y - \mu) = E(Y) - \mu = 0$$

- If you want to consider the size of the deviation, you either need to consider the absolute value or the square of $Y - \mu$. In algebra, it is easier to use the square values than the absolute values, you need to consider $E[(Y - \mu)^2]$, then take the square root to get the value in the units same as Y

$$E[(Y - \mu)^2] = E[Y^2 - 2\mu Y + \mu^2]$$

$$Var\ (Y) = E(Y^2) - 2\mu^2 + \mu^2 \text{ because } E(Y) = \mu$$

$$= E(Y^2) - \mu^2 = E(Y^2) - [E(Y)]^2$$

Example

- Let Y be a discrete random variable with probability function as given below

y_i	$f(y_i)$
0	0.20
1	0.15
2	0.25
3	0.35
4	0.05

- **Expected Value**
- $E(Y) = 0 \times 0.20 + 1 \times 0.15 + 2 \times 0.25 + 3 \times 0.35 + 4 \times 0.05 = 1.90$

Example

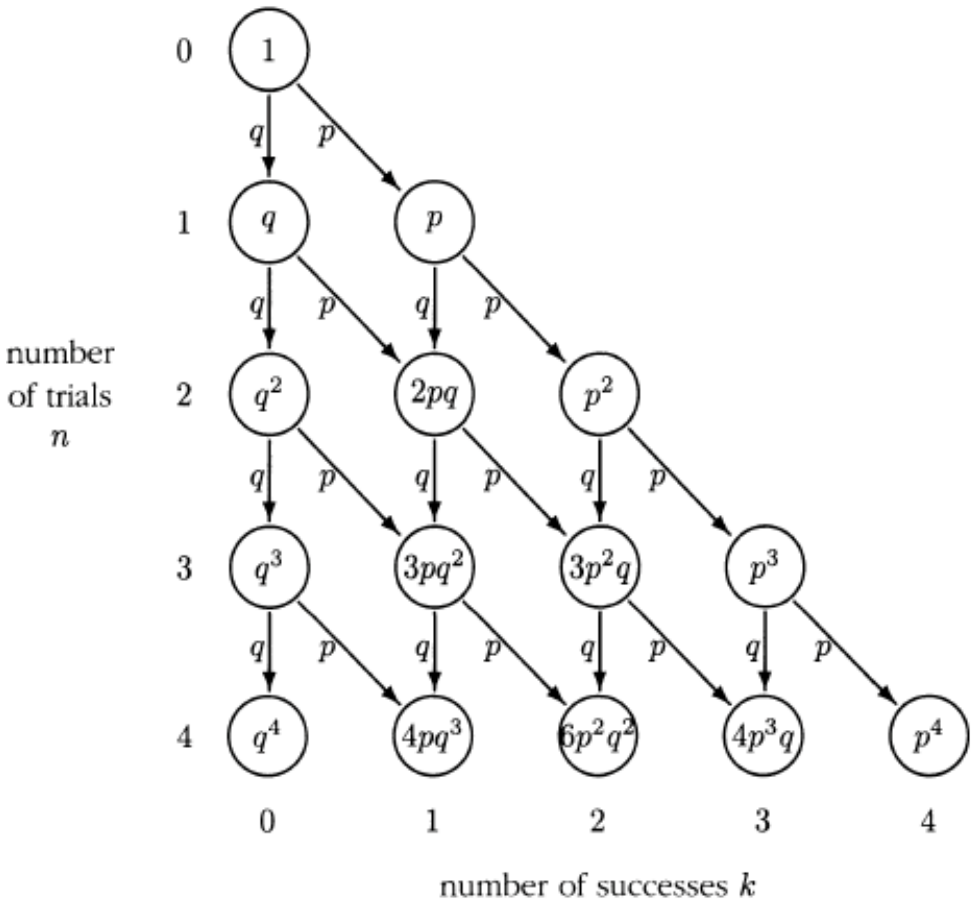
y_i	$f(y_i)$
0	0.20
1	0.15
2	0.25
3	0.35
4	0.05

- **Variance**
- Variance can be calculated in two way
- $Var\ (Y) = (0 - 1.90)^2 \times 0.20 + (1 - 1.90)^2 \times 0.15 + (2 - 1.90)^2 \times 0.25 + (3 - 1.90)^2 \times 0.35 + (4 - 1.90)^2 \times 0.05 = 1.49$
- Second way is
- $E(Y^2) = 0^2 \times 0.20 + 1^2 \times 0.15 + 2^2 \times 0.25 + 3^2 \times 0.35 + 4^2 \times 0.05 = 5.10$
- $Var\ (Y) = 5.10 - 1.90^2 = 1.49$

Binomial Distribution

Binomial Distribution

- We need to find a formula for finding the probability of getting k successes in n independent trials.
- For this, we consider a tree diagram for $n = 4$
- Each path down the n steps represents the possible outcomes of the first n trials.
- The kth node in the nth trial represents k successes in n trials.

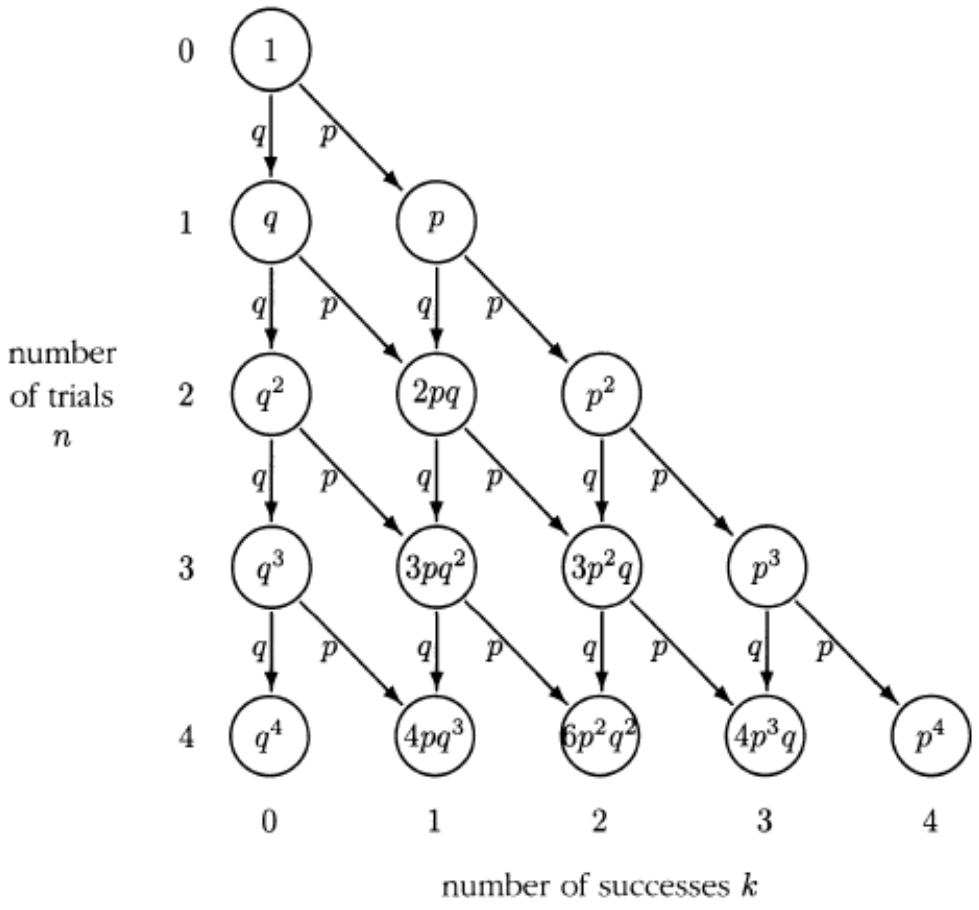


Samatrix.io

Binomial Distribution

- The expression in each node denotes the probabilities of success (denoted by p) and failure (denoted by $1 - p = q$) on each trial.
- The expression shows the sum of probabilities of all paths leading to the node.
- For example in row 3, the probabilities of $k = 0, 1, 2, 3$ successes in $n = 3$ can be expressed by

$$(p + q)^3 = q^3 + 3pq^2 + 3p^2q + p^3$$



Samatrix.io

Binomial Distribution

- The second term on the right-hand side $3pq^2$ denotes the 1 success from 3 trials $(k = 1, n = 3)$.
- The factor 3 arises because there are three ways to get one success in three trials. FFS, FSF, SFF.
- It also represents the three possible ways to reach the first node in the row 3.
- From n trials if we want to achieve k successes, we should move down to the right k times, that is corresponding to k successes and straight down $n - k$ times that is corresponding to $n - k$ failures.
- The probability of every path of k successes in the n trials is $p^k q^{n-k}$.
- So, the probability mass function of all the paths of k successes in the n trials is

$$f(k) = \binom{n}{k} p^k q^{n-k}$$

Binomial Distribution

- Where as $\binom{n}{k}$ denotes number of paths. It is called $n\ choose\ k$. The term $\binom{n}{k}$ is given by the formula

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1} = \frac{n!}{k!\,(n-k)!}$$

- The n and p are fixed where as k varies.
- The binomial probabilities thus define a probability distribution known as Binomial Probability (n, p) distribution.
- The binomial distribution is a distribution of number of successes in n independent and identical trials with the probability of success for each trial as p.

Example

Calculate the probability mass function of getting 2 fives and 7 non-fives in 9 rolls of a dice

$$\binom{9}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^7 = \frac{36 \times 5^7}{6^9} = 0.279$$

We can solve this problem by using scipy.stats.binom.pmf() function.

```
>>> from scipy.stats import binom
>>> k,n,p=2,9,1/6
>>> binom.pmf(k,n,p)
0.2790816472336532
```

Example2: In the case of fair coin tossing: $p = q = 1/2$ so

$$p^k q^{n-k} = \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k} = \left(\frac{1}{2}\right)^n$$

- Probability of k heads in n fair coin tosses = $\binom{n}{k} \times \frac{1}{2^n}$ where as $(0 \leq k \leq n)$

Create function to calculate probability mass function for binary events.

Properties of Binomial Distribution

- There are n independent and identical trials for a Bernoulli (success-failure) experiment.
- There are two possible outcomes of every experiment: Success or failure
- The probability of success (denoted by p) remains constant for each trial
- k is a random variable that denotes the number of “successes” observed during n trials.

Mean and Standard Deviation

- Like any other probability distributions, a binomial probability distribution also has a mean μ and standard deviation, σ.

$$\mu = np$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{npq}$$

Mean and Standard Deviation

- Example: A company that is producing turf grass monitors the quality of grass by taking a sample on 25 seeds on a regular interval. The germination rate of the seed is consistent at 85%. Find the average number of seeds that will germinate in the sample of 25 seeds
- $\mu = np = 25 \times 0.85 = 21.25$
- $\sigma = \sqrt{25 \times 0.85 \times 0.15} = 1.785$

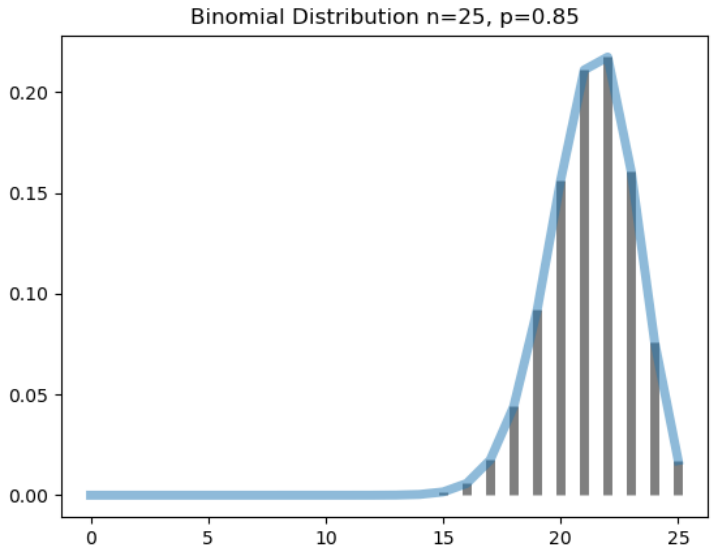
- **Using Python**

```
>>> from scipy.stats import binom
>>> n, p = 25, 0.85
>>> binom.mean(n,p)
21.25
>>> binom.std(n,p)
1.7853571071357126
```

Binomial Probability Distribution

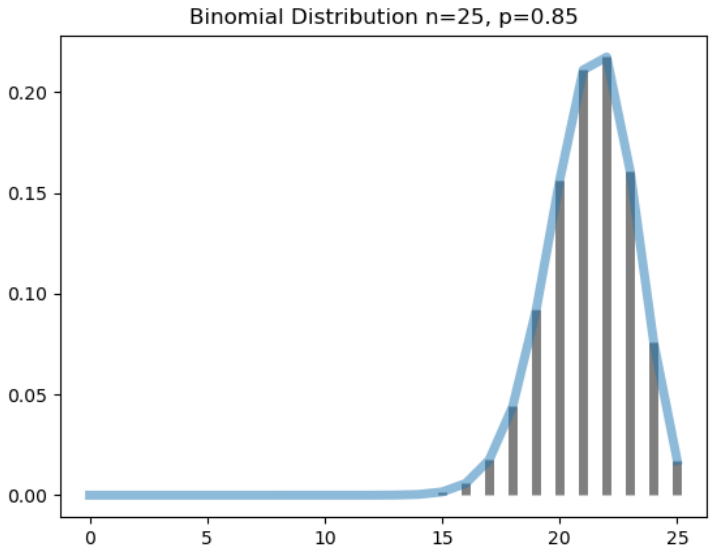
- We can plot the probability distribution whereas the distribution is tending towards left skewness

```
>>> import numpy as np
>>> from scipy.stats import binom
>>> import matplotlib.pyplot as plt
>>> x = np.arange(0,26)
>>> n, p = 25, 0.85
>>> fig, ax = plt.subplots(1, 1)
>>> ax.plot(x,binom.pmf(x,n,p),lw=5,alpha=0.5)
>>> ax.vlines(x,0,binom.pmf(x,n,p),lw=5,alpha=0.5)
>>> plt.title('Binomial Distribution n=25, p=0.85')
>>> plt.show()
```



Binomial Probability Distribution

- The bimonial(n, p) distributions have roughly the same bell shape irrespective of values of n and p. As n and p vary, binomial distributions differ in their mean and standard deviation

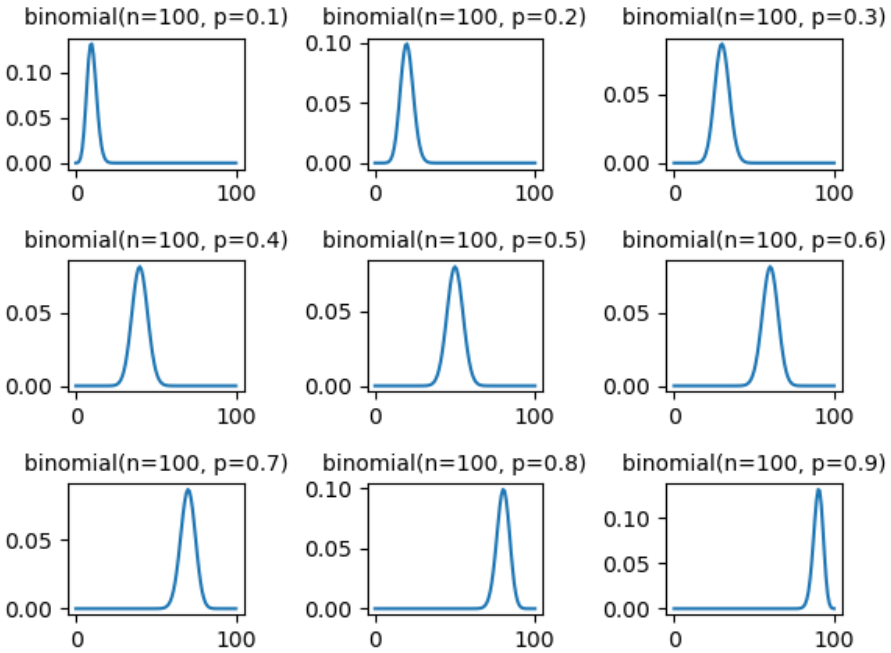


Samatrix.io

Distribution of number of success for n trials

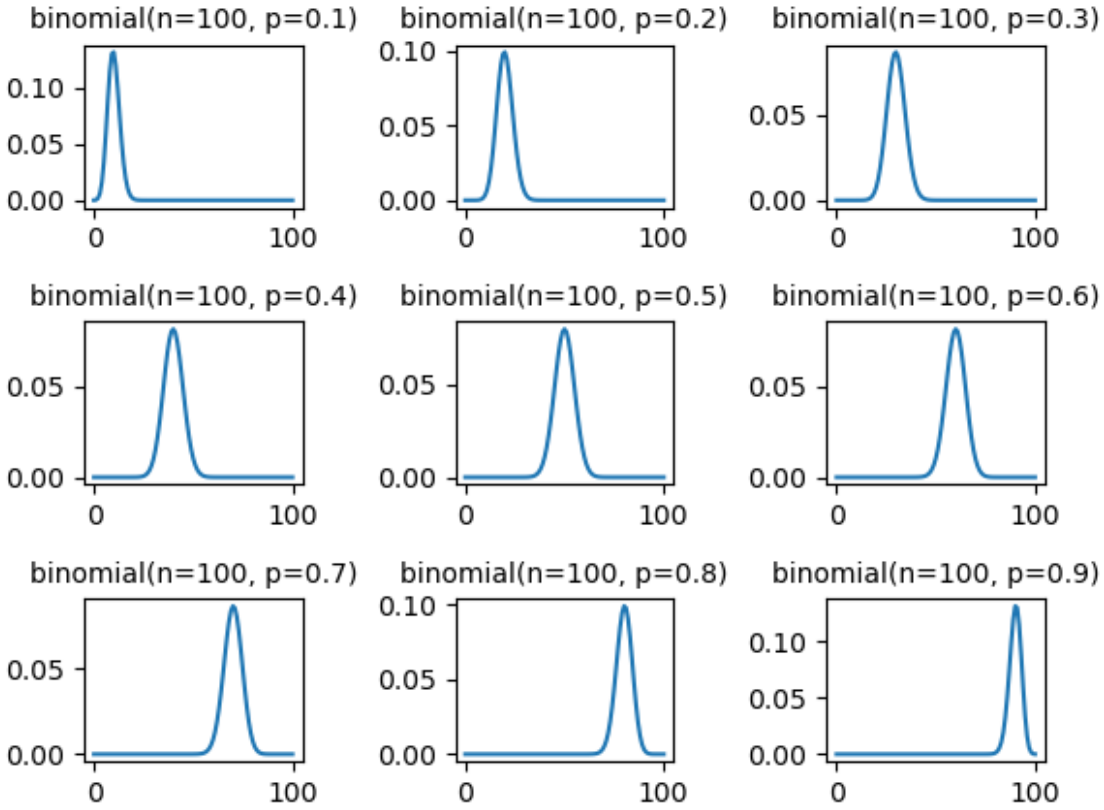
- The bimonial$(100, p)$ distribution is shown for $p = 10\%\ to\ 90\%$ by step of 10%. With an increase in p the distributions shifts to the right because the distribution is centered around mean, $100p$, which increases with p. The distribution is symmetric around $p = 0.5$ and skewed around $p = 0\ and\ p = 1$. The spread of the distribution increases with p till 50% where it is maximum and then start reducing. This is justified from the formula of the standard deviation $\sqrt{np(1-p)}$ that increases with value of p till 50% and then reduces.
-

```
import numpy as np
from scipy.stats import binom
import matplotlib.pyplot as plt
fig, ((ax1, ax2, ax3), (ax4, ax5, ax6), (ax7, ax8, ax9)) = plt.subplots(3, 3)
x=np.arange(0,101)
ax1.plot(x,binom.pmf(x,n=100,p=0.1))
ax2.plot(x,binom.pmf(x,n=100,p=0.2))
ax3.plot(x,binom.pmf(x,n=100,p=0.3))
ax4.plot(x,binom.pmf(x,n=100,p=0.4))
ax5.plot(x,binom.pmf(x,n=100,p=0.5))
```



Distribution of number of success for n trials

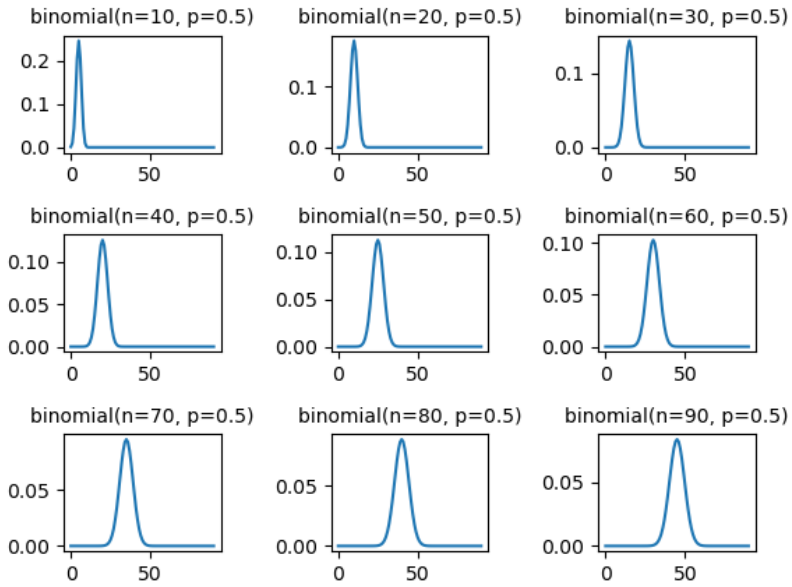
```
ax6.plot(x,binom.pmf(x,n=100,p=0.6))
ax7.plot(x,binom.pmf(x,n=100,p=0.7))
ax8.plot(x,binom.pmf(x,n=100,p=0.8))
ax9.plot(x,binom.pmf(x,n=100,p=0.9))
ax1.set_title("binomial(n=100, p=0.1)",fontsize=10)
ax2.set_title("binomial(n=100, p=0.2)",fontsize=10)
ax3.set_title("binomial(n=100, p=0.3)",fontsize=10)
ax4.set_title("binomial(n=100, p=0.4)",fontsize=10)
ax5.set_title("binomial(n=100, p=0.5)",fontsize=10)
ax6.set_title("binomial(n=100, p=0.6)",fontsize=10)
ax7.set_title("binomial(n=100, p=0.7)",fontsize=10)
ax8.set_title("binomial(n=100, p=0.8)",fontsize=10)
ax9.set_title("binomial(n=100, p=0.9)",fontsize=10)
fig.subplots_adjust(hspace=0.7)
fig.subplots_adjust(wspace=0.7)
plt.show()
```



Distribution of Number of Heads for n coin tosses

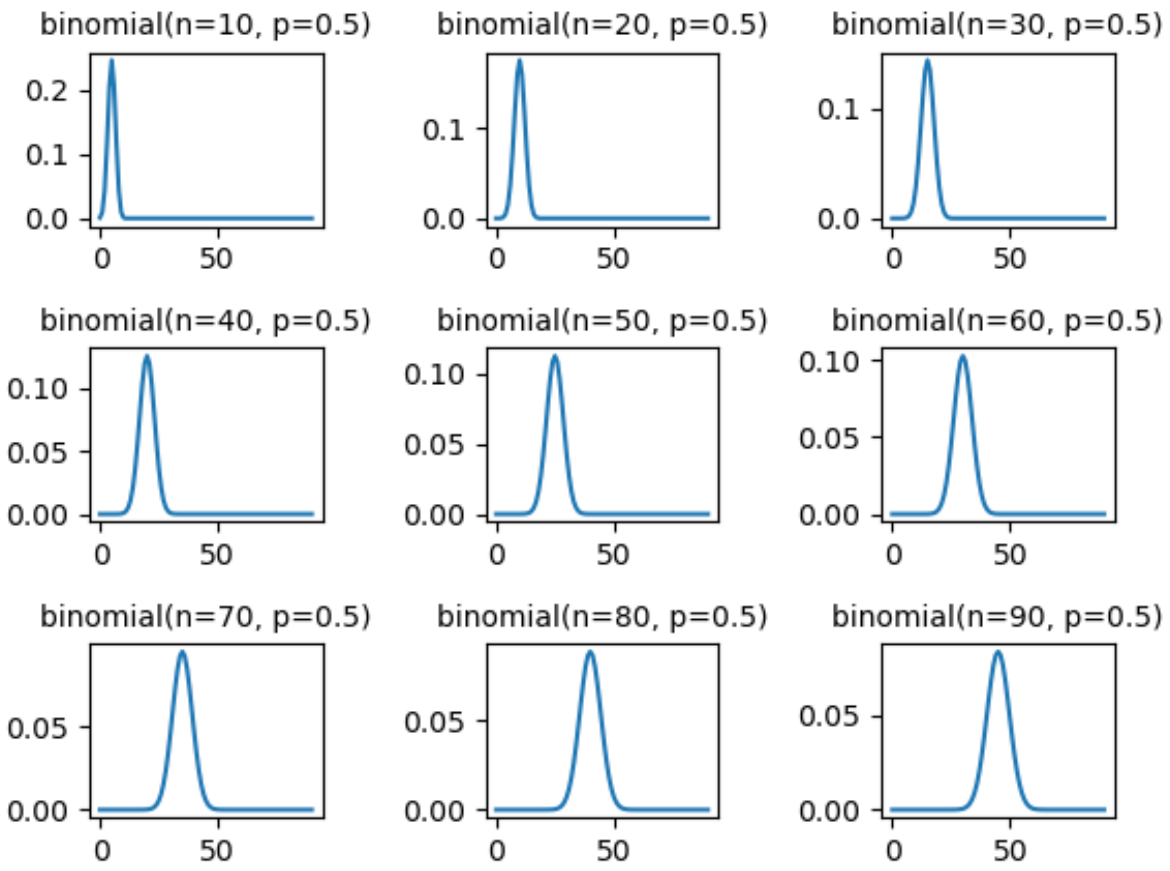
- The bimonial$(n, 0.5)$ distribution is shown for $n = 10\ to\ 90$ by step of 10. With an increase in n the distributions shifts to the right because the distribution is centered around mean, $n/2$, which increases with p. The distribution is symmetric around the expected value $n/2$. The spread of the distribution increases with n. This is justified from the formula of the standard deviation $\sqrt{np(1-p)}$ that increases with value of n. Due to increase in spread the distribution covers a wider range of values.

```
import numpy as np
from scipy.stats import binom
import matplotlib.pyplot as plt
fig, ((ax1, ax2, ax3),(ax4, ax5, ax6),
(ax7, ax8, ax9)) = plt.subplots(3,3)
x=np.arange(0,91)
ax1.plot(x,binom.pmf(x,n=10,p=0.5))
ax2.plot(x,binom.pmf(x,n=20,p=0.5))
ax3.plot(x,binom.pmf(x,n=30,p=0.5))
ax4.plot(x,binom.pmf(x,n=40,p=0.5))
```



Distribution of Number of Heads for n coin tosses

```
ax5.plot(x,binom.pmf(x,n=50,p=0.5))
ax6.plot(x,binom.pmf(x,n=60,p=0.5))
ax7.plot(x,binom.pmf(x,n=70,p=0.5))
ax8.plot(x,binom.pmf(x,n=80,p=0.5))
ax9.plot(x,binom.pmf(x,n=90,p=0.5))
ax1.set_title("binomial(n=10, p=0.5)",fontsize=10)
ax2.set_title("binomial(n=20, p=0.5)",fontsize=10)
ax3.set_title("binomial(n=30, p=0.5)",fontsize=10)
ax4.set_title("binomial(n=40, p=0.5)",fontsize=10)
ax5.set_title("binomial(n=50, p=0.5)",fontsize=10)
ax6.set_title("binomial(n=60, p=0.5)",fontsize=10)
ax7.set_title("binomial(n=70, p=0.5)",fontsize=10)
ax8.set_title("binomial(n=80, p=0.5)",fontsize=10)
ax9.set_title("binomial(n=90, p=0.5)",fontsize=10)
fig.subplots_adjust(hspace=0.7)
fig.subplots_adjust(wspace=0.7)
plt.show()
```



Example

For the US presidential elections, there are 4 races. In each race, Republicans have 60% chances of winning. If each race is independent of each other, what is the probability that

- The Republicans will win 0 races, 1 race, 2 race, 3 race or all the 4 races
- The Republicans will win at least one race
- The Republicans will win the majority of the races

Let X equals the number of races

- $\binom{4}{0}p^0q^4 = \frac{4!}{0!(4-0)!}0.6^0 0.4^4 = 0.4^4 = 0.0256$
- $\binom{4}{1}p^1q^3 = \frac{4!}{1!(4-1)!}0.6^1 0.4^3 = 4 \times 0.6^1 \times 0.4^3 = 0.1536$
- $\binom{4}{2}p^2q^2 = \frac{4!}{2!(4-2)!}0.6^2 0.4^2 = 6 \times 0.6^2 \times 0.4^2 = 0.3456$
- $\binom{4}{3}p^3q^1 = \frac{4!}{3!(4-3)!}0.6^3 0.4^1 = 4 \times 0.6^3 \times 0.4^1 = 0.3456$
- $\binom{4}{4}p^4q^0 = \frac{4!}{4!(4-4)!}0.6^4 0.4^0 = 0.6^4 \times 0.4^0 = 0.129$

- b. $P(\text{at least } 1) = 1 - P(0) = 0.9744$ or $P(1) + P(2) + P(3) + P(4) = 0.9744$
- c. $P(\text{Republicans wins the majority}) = P(3) + P(4) = 0.4752$

Example – Using Python

```
>>> from scipy.stats import binom
>>> binom.pmf(k=0, n=4, p=0.6)
0.025600000000000008
>>> binom.pmf(k=1, n=4, p=0.6)
0.15360000000000007
>>> binom.pmf(k=2, n=4, p=0.6)
0.3456000000000001
>>> binom.pmf(k=3, n=4, p=0.6)
0.3456000000000001
>>> binom.pmf(k=4, n=4, p=0.6)
0.1296
>>> 1-binom.cdf(k=0,n=4,p=0.6)
0.9744
>>> 1- binom.cdf(k=2,n=4,p=0.6)
0.47520000000000007
```

Example

- We use 1- binom.cdf(k=2,n=4,p=0.6) to check the probability of 3 success or more whereas binom.cdf(k=2,n=4,p=0.6) to check the probability of 2 successes or less.

```
>>> binom.cdf(k=2,n=4,p=0.6)
0.5247999999999999
```

- This example demonstrates the difference between cumulative distribution function, cdf() and Probability mass function, pmf() functions.
- Probability mass function is used to check mass (proportion of observations) at a given number of successes k. Cumulative density function is used to check the probability of achieving the successes within a certain range.

- Each sample of water has a 10% chance of containing a particular organic pollutant. Assume that the samples are independent with regard to the presence of the pollutant.

1) Find the probability that in the next 18 samples, exactly 2 contain the pollutant.

2) Determine the probability that at least four samples contain the pollutant.

- The random variable X has a binomial distribution with $n = 10$ and $p = 0.5$. Determine the following probabilities:

(a) $P(X = 5)$ (b) $P(X \leq 2)$

(c) $P(X \geq 9)$ (d) $P(3 \leq X < 5)$

Samatrix.io

- A bank has issued 200 personal loans to its customers. Based on past data, the probability of a customer defaulting on a loan is estimated to be 0.05 (5%). The bank wants to assess the potential financial risk involved by calculating the probability of a certain number of loan defaults.
- **Tasks:**

1. Calculate the probability that exactly 10 customers will default on their loans out of the 200 loans issued.
2. Determine the probability that no more than 5 customers will default on their loans.
3. Assess the probability that 15 or more customers will default on their loans.

Poisson Distribution

Poisson Distribution

- Poisson distribution is another distribution for discrete random variables. When n is large and p is very close to 0 or 1, the binomial distribution is not even approximately symmetric. When p is close to 0, q is close to 1. In such case the standard deviation $\sigma = \sqrt{npq}$ is

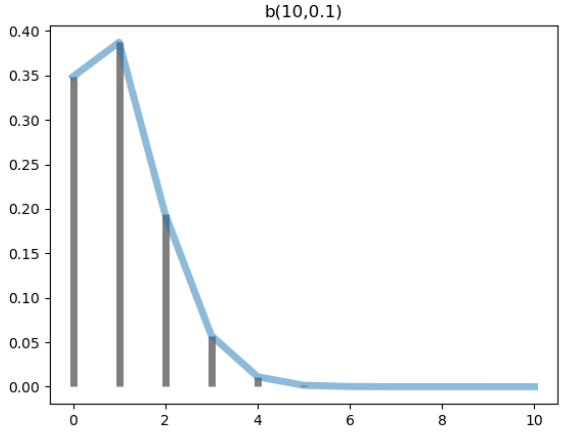
$$\sigma = \sqrt{\mu q} \approx \sqrt{\mu}$$

- Where $\mu = np$ is the mean. If we consider n trials with probability of success $p = \frac{1}{n}$, in such case $\mu = 1$ and $\sigma \approx 1$. This leads to bad normal approximation irrespective of a larger value of n.

Example - The binomial (10,0.1) distribution

A box contains 1 red ball and 9 white balls. The distribution of number of red ball picked in 10 random draws with replacement is as follows

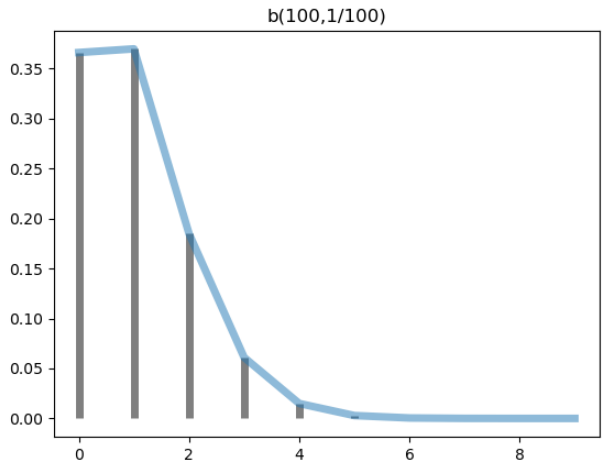
```
>>> import numpy as np
>>> from scipy.stats import binom
>>> import matplotlib.pyplot as plt
>>> x = np.arange(0,11)
>>> n, p = 10, 0.1
>>> fig, ax = plt.subplots(1, 1)
>>> ax.plot(x,binom.pmf(x,n,p),lw=5,alpha=0.5)
>>> ax.vlines(x,0,binom.pmf(x,n,p),lw=5,alpha=0.5)
>>> plt.title('b(10,0.1)')
>>> plt.show()
```



Samatrix.io

binomial (100,0.01) distribution

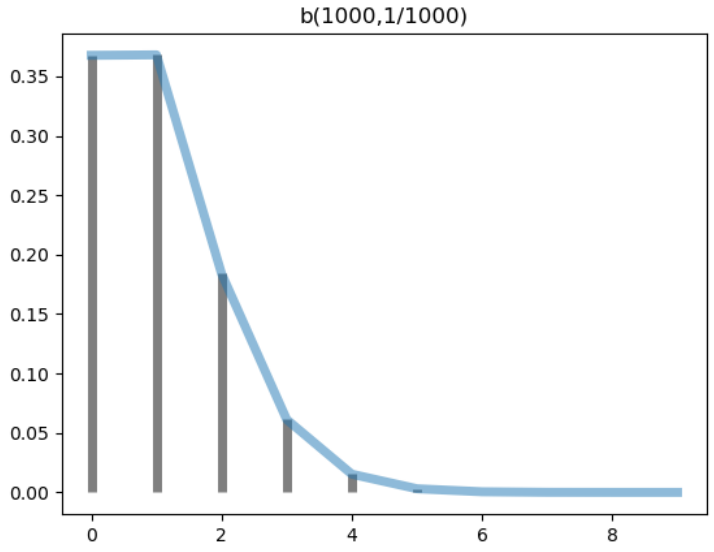
A box contains 1 red ball and 99 white balls. The distribution of number of red ball picked in 100 random draws with replacement is as follows




```
>>> x = np.arange(0,10)
>>> n, p = 100, 0.01
>>> fig, ax = plt.subplots(1, 1)
>>> ax.plot(x,binom.pmf(x,n,p),lw=5,alpha=0.5)
>>> ax.vlines(x,0,binom.pmf(x,n,p),lw=5,alpha=0.5)
>>> plt.title('b(100,1/100)')
>>> plt.show()
```

Binomial (1000,0.001) distribution

- A box contains 1 red ball and 999 white balls. The distribution of number of red ball picked in 1000 random draws with replacement is as follows



Samatrix.io

Define – Poisson Distribution

- It can be shown that the binomial distributions are always concentrated around a smaller number of values with mean value $\mu = 1$. The shape of the distribution approaches a limit with $n \to \infty$ and $p = \frac{1}{n} \to 0$
- When expected value $\mu = np$ is constant and the binomial(n, p) approaches the limits $n \to \infty$ and $p = \frac{1}{n} \to 0$, we get the Poisson distribution with parameter μ.
- So, if n is large and p is small, the distribution of success of independent n trials depend on value of $\mu = np$. The Poisson approximation states

$$P(k \text{ success}) = e^{-\mu}\left(\frac{\mu^k}{k!}\right)$$

- The Poisson distribution with parameter μ or Poisson (μ) distribution is defined as the distribution of probabilities $P_\mu(k) = e^{-\mu}\left(\frac{\mu^k}{k!}\right)$ Where as $k = 0,1,2, \ldots$

Example

- A manufacturing process produces 1% defective items in a long run. What is the probability of getting 2 or more defective samples in sample of 200 items?
- Mean $\mu = 200 \times 0.01 = 2$. We can use Poisson approximation

$$P(2 \text{ or more defectives}) = 1 - P(0) - P(1)$$

$$= 1 - e^{-2}\frac{2^0}{0!} - e^{-2}\frac{2^1}{1!} = 1 - 3e^{-2} = 0.594$$

```
>>> from scipy.stats import poisson
>>> mu = 2
>>> 1 - poisson.pmf(0, mu) - poisson.pmf(1, mu)
0.5939941502901619
>>> 1 - poisson.cdf(1,mu)
0.593994150290162
```

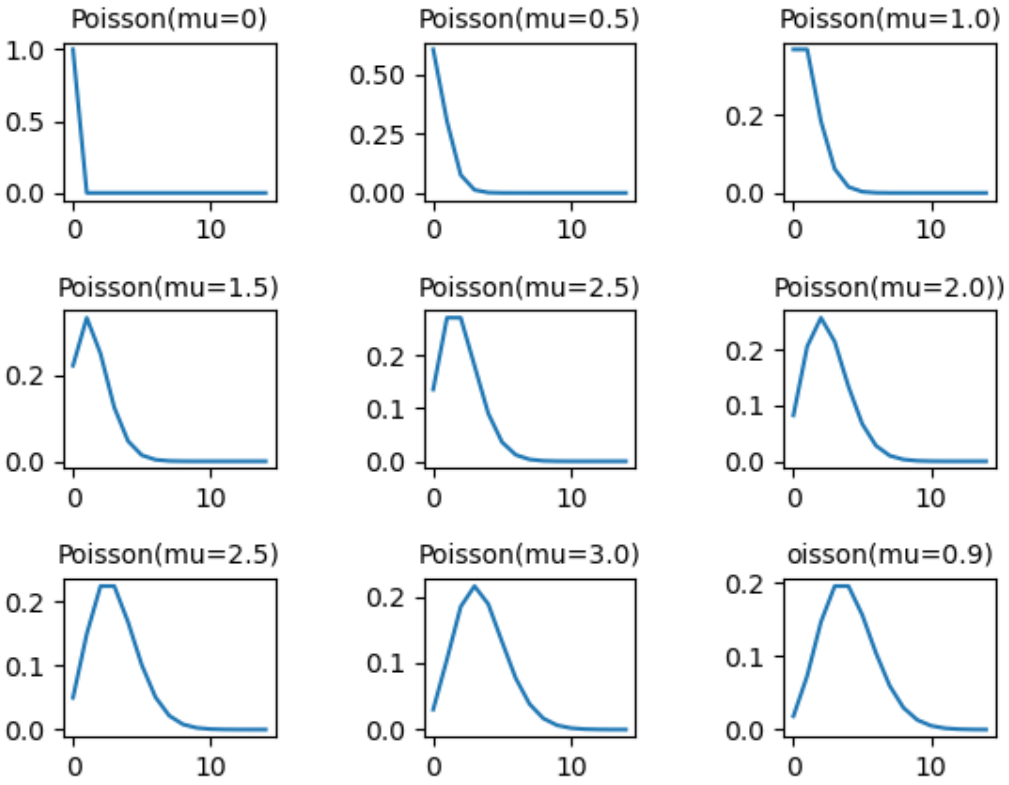
- For the case of the thin copper wire, suppose that the number of flaws follows a Poisson distribution with a mean of 2.3 flaws per millimeter. Determine the probability of exactly two flaws in 1 millimeter of wire.

- Determine the probability of 10 flaws in 5 millimeters of wire. Let X denote the number of flaws in 5 millimeters of wire.

Samatrix.io

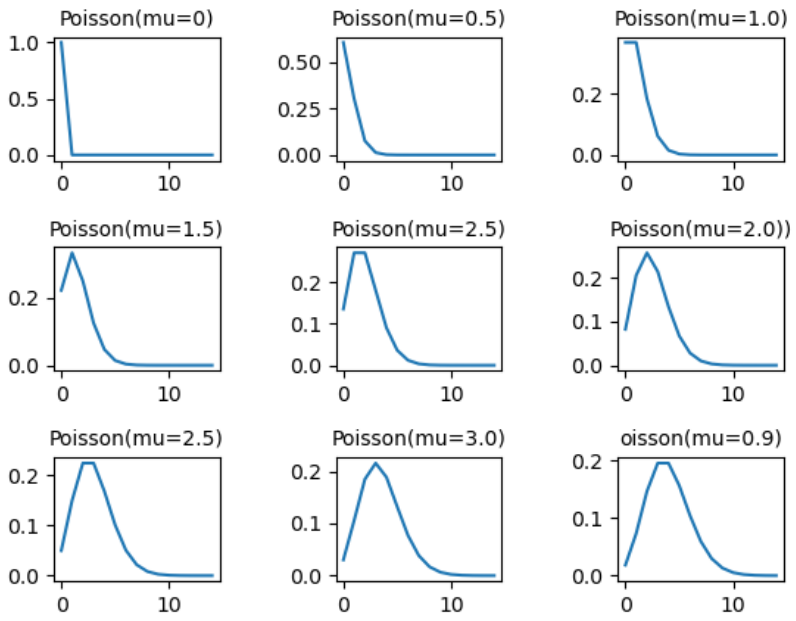
Properties of Poisson Distribution

- For small value of μ, the distribution is piled up near zero. With an increase in μ, the distribution shifts to right and spreads out. As $\mu \to \infty$, the distribution approaches the normal distribution.



Properties of Poisson Distribution

```
import numpy as np
from scipy.stats import poisson
import matplotlib.pyplot as plt
fig, ((ax1, ax2, ax3), (ax4, ax5, ax6), (ax7, ax8, ax9)) = plt.subplots(3, 3)
x=np.arange(0,15)
ax1.plot(x,poisson.pmf(x,mu=0))
ax2.plot(x,poisson.pmf(x,mu=0.5))
ax3.plot(x,poisson.pmf(x,mu=1.0))
ax4.plot(x,poisson.pmf(x,mu=1.5))
ax5.plot(x,poisson.pmf(x,mu=2.0))
ax6.plot(x,poisson.pmf(x,mu=2.5))
ax7.plot(x,poisson.pmf(x,mu=3.0))
ax8.plot(x,poisson.pmf(x,mu=3.5))
ax9.plot(x,poisson.pmf(x,mu=4.0))
```



Samatrix.io

Properties of Poisson Distribution

```
ax1.set_title("Poisson(mu=0)",fontsize=10)
ax2.set_title("Poisson(mu=0.5)",fontsize=10)
ax3.set_title("Poisson(mu=1.0)",fontsize=10)
ax4.set_title("Poisson(mu=1.5)",fontsize=10)
ax5.set_title("Poisson(mu=2.5)",fontsize=10)
ax6.set_title("Poisson(mu=2.0))",fontsize=10)
ax7.set_title("Poisson(mu=2.5)",fontsize=10)
ax8.set_title("Poisson(mu=3.0)",fontsize=10)
ax9.set_title(”Poisson(mu=0.9)",fontsize=10)
fig.subplots_adjust(hspace=0.7)
fig.subplots_adjust(wspace=0.7)
plt.show()
```

Mean and Variance

- For Poisson (μ) distribution

$$E(y|\mu) = \mu$$

$$Var(y|\mu) = \mu$$

$$\sigma(y|\mu) = \sqrt{\mu}$$

Mean and variance of the Poisson (μ) are equal to μ

Samatrix.io

Characteristics of Poisson Distribution

Poisson distribution is a limiting case of a binomial distribution. For a binomial distribution if $n \to \infty$ and $p \to 0$ and the expected value $\mu = np$ remains constant we get Poisson approximation.

- In the binomial distribution, the probability of success is constant across all the trials. For the Poisson, the instantaneous rate of occurrences per unit time or space is constant. It means that the expected number of events during a given time period is same as number of events during some other time period at a later stage.
- In case of binomial the trials are independent. The occurrences of Poisson events in two non-overlapping intervals will be independent of each other. The Poisson events occur randomly throughout the given period of time at constant instantaneous rate.

Characteristics of Poisson Distribution

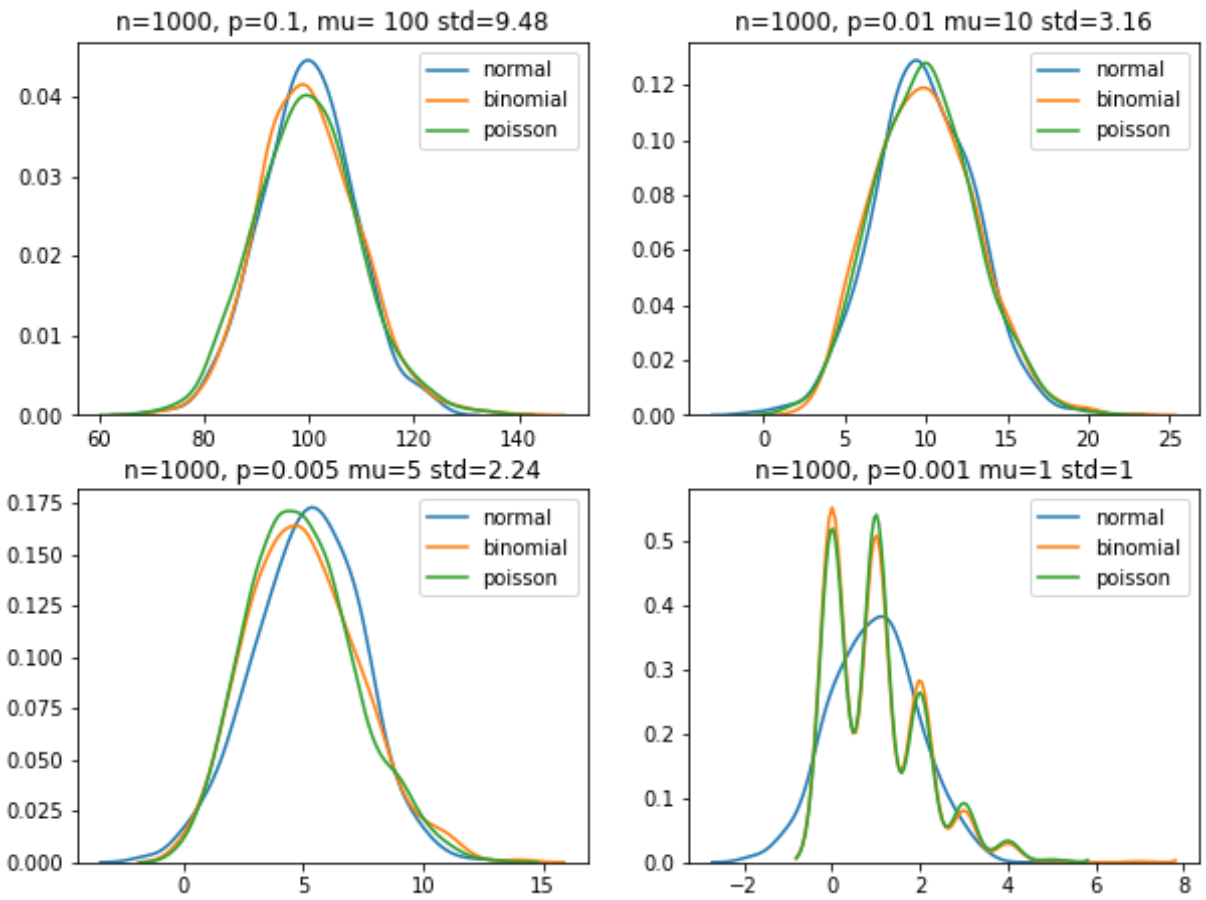
- In case of binomial, the possible outcome of an event is either success or failure. The Poisson event occur one at a time

Even though these conditions seem to be restrictive, many real-life situations satisfy these conditions. For example, number of arrivals at the ticket counter, bank teller counter, parking lot payment counter, or toll booth during a given period of time such as one minute could be approximated using the Poisson probability distribution.

Binomial vs Poisson vs Normal Distribution

- Even though it is very easy to calculate the Binomial coefficient but for large values of n, the computation of binomial coefficient becomes challenging.
- We can use the binomial approximation techniques to bypass such problems.
- Normal approximation to binomial works very well when the variance $np(1-p)$ is large.
- As we can see when the value of $\mu = np \geq 10$, the normal distribution approximates the binomial distribution very well.
- However for a low values of p such that $\mu = np = 1$ and $\sigma = \sqrt{np(1-p)} = 1$, the normal distribution is not able to approximate the binomial distribution.
- In this case the Poisson distribution can approximate the binomial distribution very well.

Binomial vs Poisson vs Normal Distribution



Samatrix.io

Binomial vs Poisson vs Normal Distribution

- A normal distribution is remains symmetric for all the values of μ and σ but the basic shape of the Poisson distribution changes with the value of μ.
- The Poisson distribution is highly skewed for lower values of μ.
- Most of the values are piled up near 0.
- For higher values of μ, the Poisson distribution appears to take the symmetric shape but technically it cannot.
- Because for the Poisson distribution mean = variance.

Binomial vs Poisson vs Normal Distribution

```
from numpy import random
import matplotlib.pyplot as plt
import seaborn as sns
fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(2, 2)
fig.set_size_inches(10,7.5)
x=np.arange(0,15)
sns.distplot(random.normal(loc=500, scale=15.81, size=1000), hist=False, label='normal', ax=ax1)
sns.distplot(random.binomial(n=1000, p=0.5, size=1000), hist=False, label='binomial', ax=ax1)
sns.distplot(random.poisson(lam=500, size=1000), hist=False, label='poisson', ax=ax1)

sns.distplot(random.normal(loc=100, scale=9.48, size=1000), hist=False, label='normal', ax=ax2)
sns.distplot(random.binomial(n=1000, p=0.1, size=1000), hist=False, label='binomial', ax=ax2)
sns.distplot(random.poisson(lam=100, size=1000), hist=False, label='poisson', ax=ax2)
```

Binomial vs Poisson vs Normal Distribution

```
sns.distplot(random.normal(loc=10, scale=3.16, size=1000), hist=False, label='normal', ax=ax3)
sns.distplot(random.binomial(n=1000, p=0.01, size=1000), hist=False, label='binomial', ax=ax3)
sns.distplot(random.poisson(lam=10, size=1000), hist=False, label='poisson', ax=ax3)

sns.distplot(random.normal(loc=1, scale=1, size=1000), hist=False, label='normal', ax=ax4)
sns.distplot(random.binomial(n=1000, p=0.001, size=1000), hist=False, label='binomial', ax=ax4)
sns.distplot(random.poisson(lam=1, size=1000), hist=False, label='poisson', ax=ax4)

ax1.set_title("n=1000, p=0.5")
ax2.set_title("n=1000, p=0.1")
ax3.set_title("n=1000, p=0.01")
ax4.set_title("n=1000, p=0.001")
plt.show()
```

Continuous Random Variable

Samatrix.io

Continuous Random Variable

- The section is about continuous random variables.
- In continuous random variables, there is an uncountably infinite number of real numbers in a given range.
- Due to which it is impossible to get the probability of a particular value, as in the case of discrete random variable.
- The probability of getting a particular value is zero.
- So, in the case of continuous random variables, we use probability density function and use calculus to compute the probabilities.

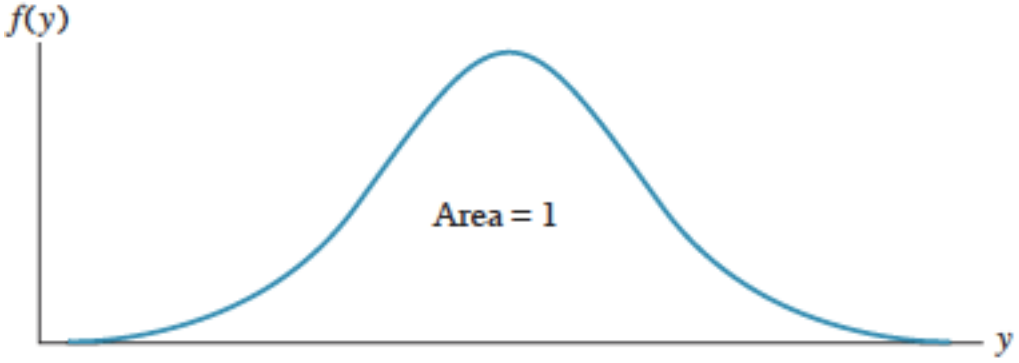
Probability Distribution Function

- In the case of histograms, we studied the concept of calculating the probability of y falling within an interval.
- As the number of the intervals goes to infinity, and the width of interval or bin goes to zero, the relative frequency histogram would almost be a smooth curve.
- This smooth curve is called the probability density function.
- The height of the probability function does not represent the probability at that point.
- Infect at every point the probability is zero. We can find how dense the probability is at a given point by measuring the height of the curve.

Probability Distribution Function

- The total area under the curve is one. Since the area under the curve can be calculated using integration

$$\int_{-\infty}^{\infty} f(y)\, dy = 1$$



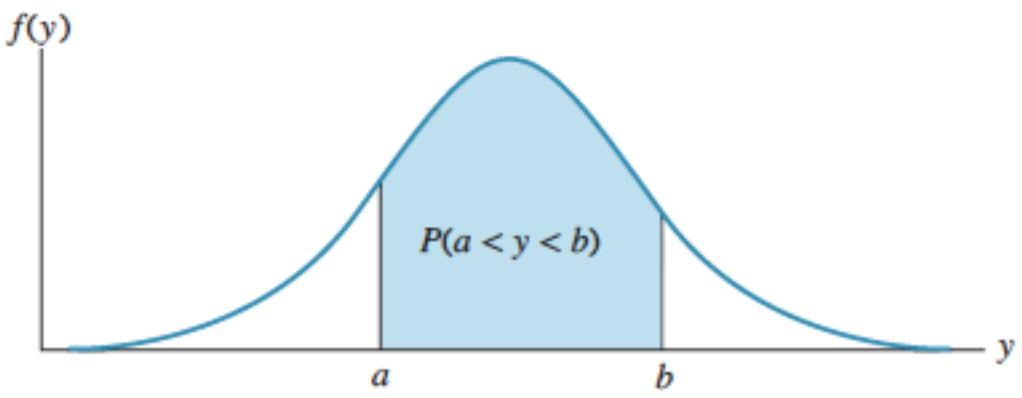
(a) Total area under the curve

Samatrix.io

Probability Distribution Function

- The area of the histogram that lies between the interval (a, b) gives the proportion of the observations that line in the interval. The proportion of the area represents the probability that the random variable falls in the interval (a, b).

$$P(a < Y < b) = \int_a^b f(y)dy$$

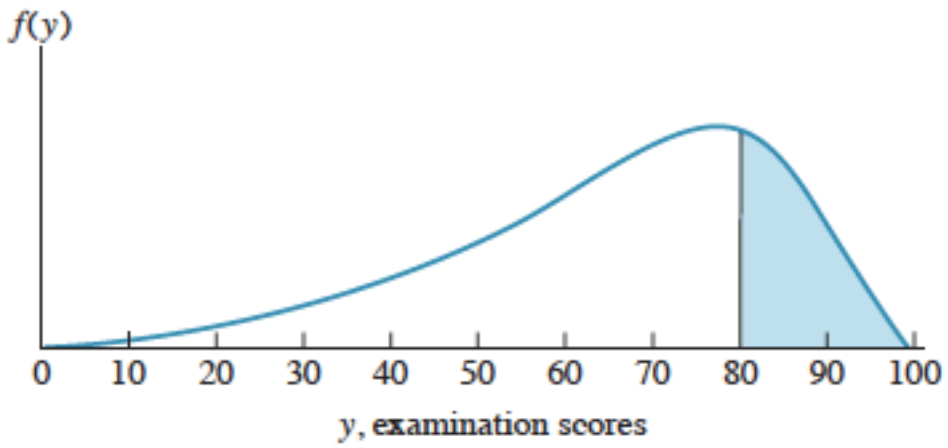


(b) Probability

Samatrix.io

Probability Distribution Function

- The relative frequency distribution curve can take several types of shapes.
- If the know the function that represents the curve, we can find out the area under the whole curve or between an interval using the integration.
- Fortunately, the functions for many of these curves are known and ready to use.
- Example: The probability distribution function of the score obtained by the students is known. We can find the probability that a particular student will score more than 80% by calculating the shaded area.



Samatrix.io

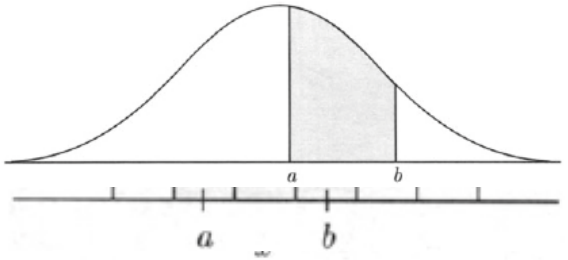
Expected Value and Variance

- For a relative frequency histogram, as the number of bars increase without bound, the width of each bar gets closer and closer to zero. In the limit, the midpoint of each bar that contain y gets closer and closer to y. The height of the bar that contains y approaches to $f(y)$. This is also known as relative frequency density. In the limit the relative frequency density gets closer to probability density. The expected value of the random variable

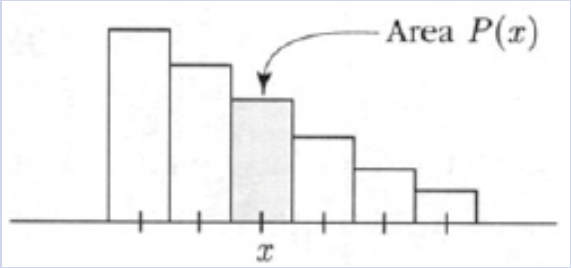
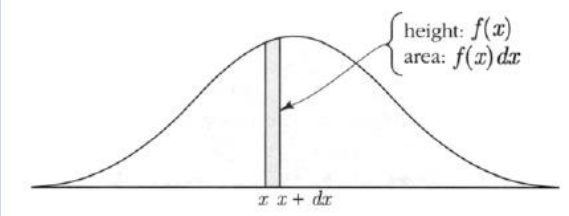
$$E(Y) = \int_{-\infty}^{\infty} y f(y) dy$$

- The expected value $E\big(Y - E(Y)\big)^2$ gives us **variance**

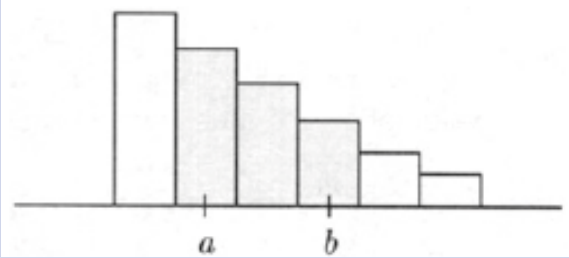
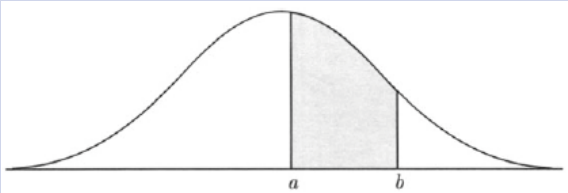
$$Var(Y) = E\big(Y - E(Y)\big)^2 = E(Y^2) - [E(Y)]^2 = \int_{-\infty}^{\infty} (y - \mu)^2 f(y) dy$$



vs Continuous Distribution

Discrete Distribution	Continuous Distribution
Point Probability  $P(X = x) = P(x)$ $P(x)$ is the probability that random variable X has integer value x	**Infinitesimal Probability**  $P(X \in dx) = f(x)dx$ The probability per unit length (density $f(x)$) for value near x

Discrete vs Continuous Distribution

Discrete Distribution	Continuous Distribution
Interval Probability  $$P(a \le X \le b) = \sum_{a \le x \le b} P(x)$$ Relative area under histogram between $a - \frac{1}{2}$ and $b + \frac{1}{2}$	**Interval Probability**  $$P(a \le X \le b) = \int_a^b f(x)dx$$ Area under the graph between a and b
Constraints Non negative with sum 1 $P(x) \ge 0$ for all x and $\sum_{all\ x} P(x) = 1$	**Constraints** Non negative with total integral 1 $f(x) \ge 0$ for all x and $\int_{-\infty}^{\infty} f(x)dx = 1$
Expectation $$E(X) = \sum_{all\ x} x \times f(x)$$	**Expectation** $$E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

Samatrix.io

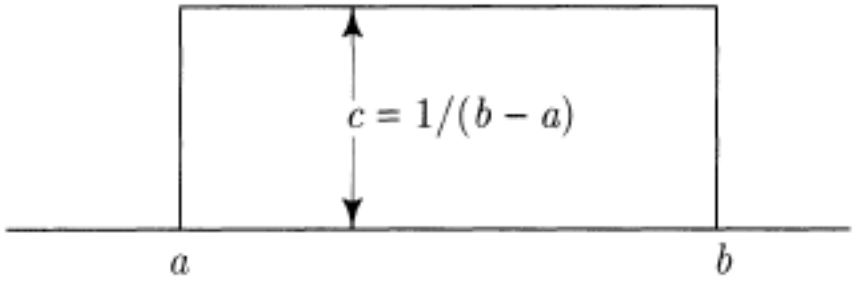
- The diameter of a particle of contamination (in micrometers) is modeled with the probability density function $f(x) = 2/(x)^3$ for $x > 1$. Determine the following:

(a) $P(X < 2)$ (b) $P(X > 5)$ (c) $P(4 < X < 8)$ (d) $P(X < 4 \text{ or } X > 8)$ (e) x such that $P(X < x) = 0.95$

Samatrix.io

Uniform Distribution

Uniform Distribution



- The random variable X has a uniform $(0,1)$ distribution if its probability density function $f(x)$ is constant on $[0,1]$ and 0 everywhere else.

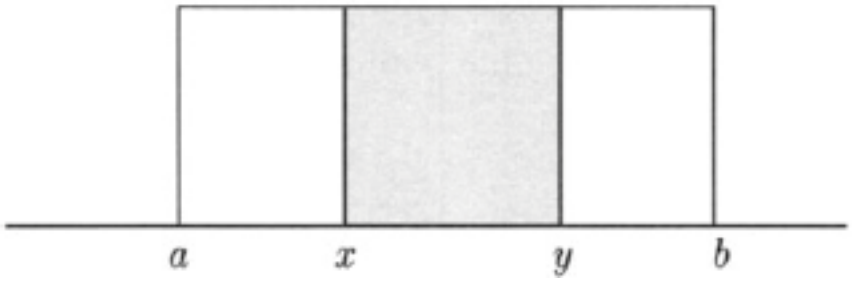
$$f(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for } \quad \notin [0,1] \end{cases}$$

- In case of uniform(a, b), the density is constant on (a, b). The value of density c is $1/(b-a)$. The total area under the density function always remains 1. So

$$(b-a)c = 1 \Longrightarrow c = 1/(b-a)$$

Samatrix.io

Uniform Distribution



- For a uniform distribution, the probabilities are relative lengths only. If X has uniform (a, b) distribution, the probability of X between x and y is

$$P(x < X < y) = \frac{\text{length } (x, y)}{\text{length } (a, b)} = \frac{y - x}{b - a}$$

- If X has a uniform $(0,2)$ distribution, the probability that X is 1.23 correct to two decimal places

$$P(1.225 < X\ 1.235) = \frac{1.235 - 1.225}{2} = \frac{0.01}{2} = 0.5\%$$

Expected Value and Variance

Expected Value

The expected value of X for the uniform distribution is given by $(a+b)/2$

For uniform$(0,1)$ distribution, the expected value of X

$$E(X) = \frac{1}{2}$$

Variance

The Variance of X can be given by $(b-a)^2/12$

For the uniform (0,1) distribution, the variance of X

$$Var(X) = \frac{1}{12}$$

- The probability density function of the length of a cutting blade is $f(x) = 1.25$ for $74.6 < x < 75.4$ millimeters.
- Determine the following:

(a) $P(X < 74.8)$ (b) $P(X < 74.8 \text{ or } X > 75.2)$

(c) If the specifications for this process are from 74.7 to 75.3

millimeters, what proportion of blades meets specifications?

Samatrix.io

- Suppose that the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of four (milliamperes)2. What is the probability that a measurement exceeds 13 milliamperes?

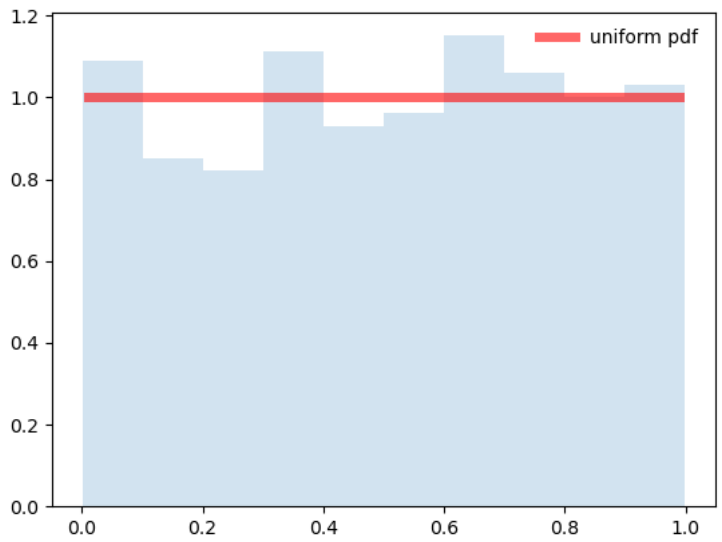
- The life of a semiconductor laser at a constant power is
- normally distributed with a mean of 7000 hours and a standard
- deviation of 600 hours.
- (a) What is the probability that a laser fails before 5000 hours?
- (b) What is the life in hours that 95% of the lasers exceed?
- (c) If three lasers are used in a product and they are assumed to
- fail independently, what is the probability that all three are
- still operating after 7000 hours?

Samatrix.io

Expected Value and Variance

```
>>> from scipy.stats import uniform
>>> uniform.mean()
0.5
>>> uniform.var()
0.08333333333333333
```

Using Python




```
from scipy.stats import uniform
import matplotlib.pyplot as plt
fig, ax = plt.subplots(1, 1)
x = np.linspace(0.01,0.99, 100)
ax.plot(x, uniform.pdf(x),'r-', lw=5, alpha=0.6, label='uniform pdf')
r = uniform.rvs(size=1000)
ax.hist(r, density=True, histtype='stepfilled', alpha=0.2)
ax.legend(loc='best', frameon=False)
plt.show()
```

Normal Distribution

Normal Distribution

- In a large population, many variables follow bell shape relative frequency distribution.
- These bell shape relative frequency distributions are symmetric and they are relatively higher in the middle than at the extremes.
- Examples of such distributions are price fluctuation of commodity in the market, stochastic aptitude test scores, physical measurement (height, weight, length) of an organism, etc.
- Each of these bell shape curve can be approximated using a normal curve.

Normal Distribution

- The normal distribution (also known as Gaussian distribution) is used to approximate a large number of probabilities distribution, so the normal distribution is the most widely used distribution in statistics.
- In general, the notation for normal distribution is $X \sim N(\mu, \sigma^2)$ identifies a normal random variable X with mean μ and variance σ^2
- The normal distribution equation for $N(\mu, \sigma^2)$ can be written as

$$y = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$$

For $-\infty < x < +\infty$.

Normal Distribution

The equation consists of two fundamental constants $\pi = 3.14159265358 \ldots$ and the base of natural algorithm $e = 2.7182818285 \ldots$

The equation of normal curve also consists of two parameters μ, the mean and σ, the standard deviation.

The equation has the term $\sqrt{2\pi}\sigma$ in denominator so that the total area under the curve is 1.

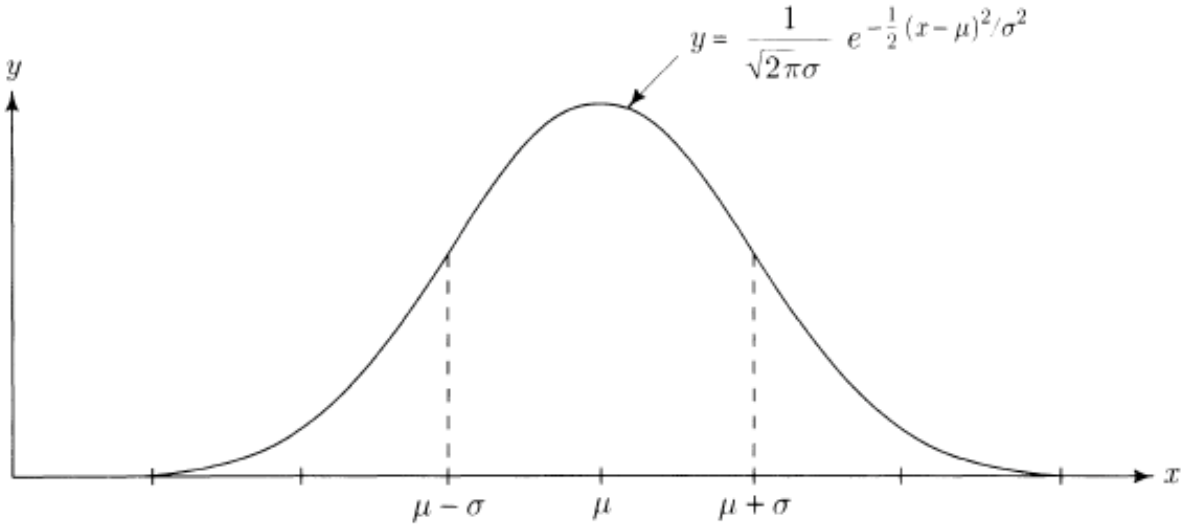
The mean, μ, can be a positive or negative real number.

The μ signifies the location of the curve.

The standard deviation, σ, can only be a positive number.

Standard deviation makes a horizontal scale and measures the spread of the distribution.

The curve is symmetric around μ. From μ to $(\mu \pm \sigma)$, the curve is concave. Beyond inflection points $\mu \pm \sigma)$, the curve becomes convex.



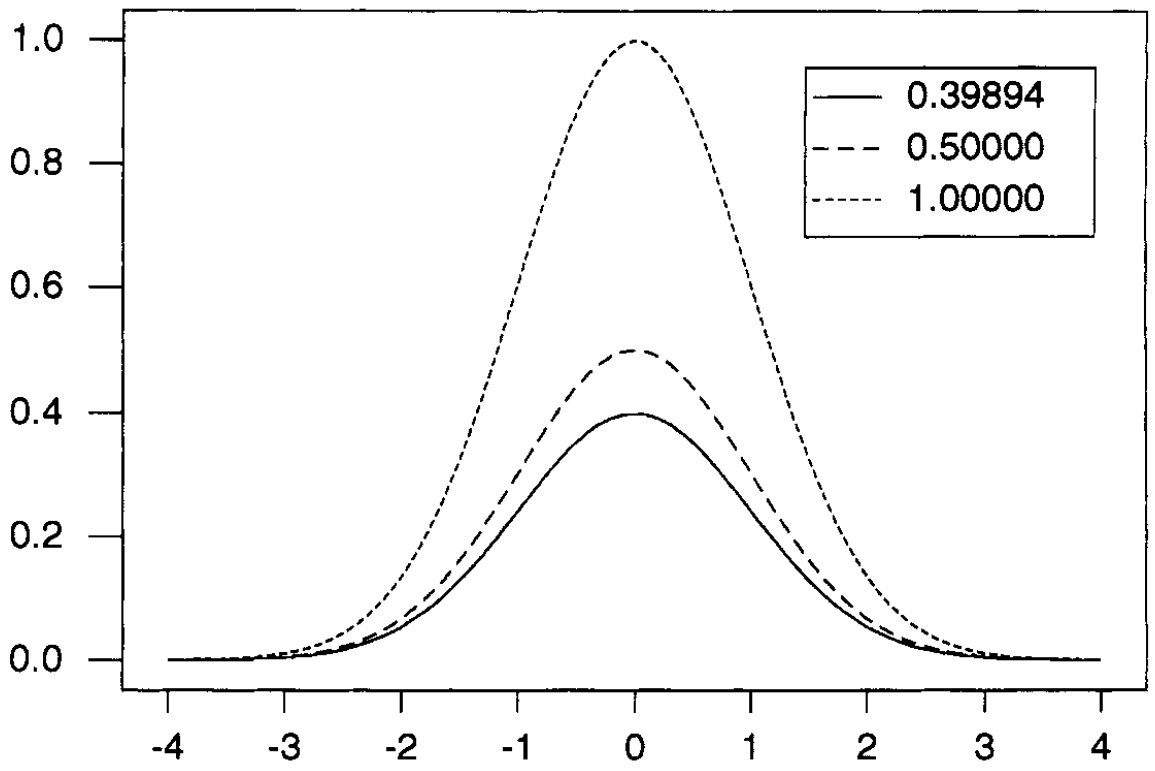
Samatrix.io

Normal Distribution Equation

- The term $e^{-\frac{1}{2}\left(\frac{(x-\mu)^2}{\sigma^2}\right)}$ determines the shape of the curve.
- Whereas the term $\frac{1}{\sqrt{2\pi}\sigma}$ does not change the basic shape of the curve.
- It just changes the area under the curve that should equate to 1.

- If we denote the term $k = \frac{1}{\sqrt{2\pi}\sigma}$, the equation becomes $ke^{-\frac{1}{2}\left(\frac{(x-\mu)^2}{\sigma^2}\right)}$.

- The curve $y = ke^{-\frac{1}{2}\left(\frac{(x-\mu)^2}{\sigma^2}\right)}$ for several values of k is as follows

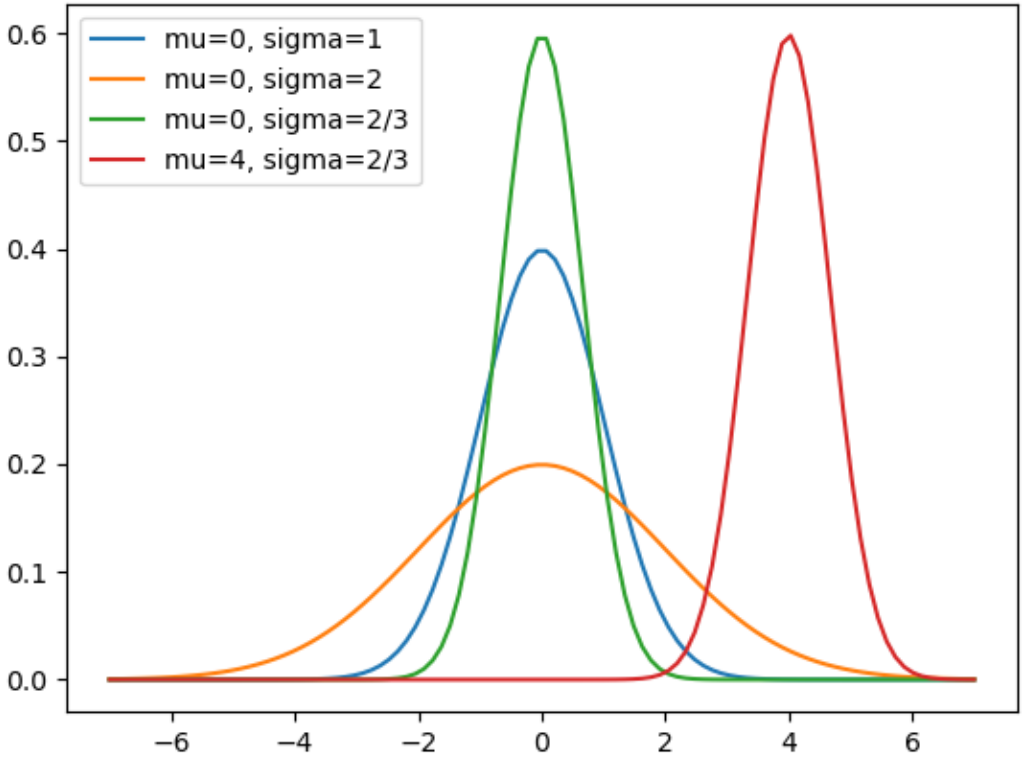


Normal Distribution Equation

- As mentioned before, the shape of the curve depends on the values of μ and σ.
- By changing the values of μ and σ, we can alter the location and the spread.
- Even by changing the values of μ and σ, we always get a bell shape curve that is mound around the mean.
- The peak of the normal distribution lies around the mean μ.

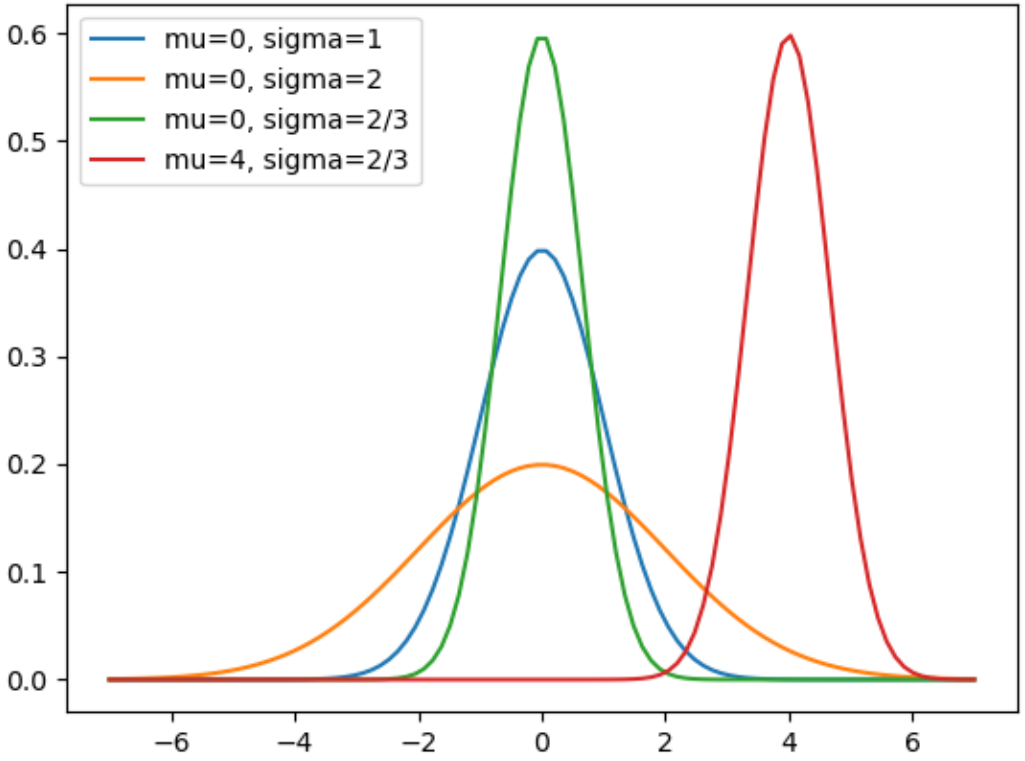
Normal Distribution

- By changing the value of μ, we can slide the distribution along the $x -$ axis.
- For example, if we increase the value of μ by 4, the whole distribution shifts to right by 4 points.
- For $\sigma^2 > 1$ and smaller values of σ, the curve is thin and tall and the values are piled up around μ.
- For higher values of σ, the values are more dispersed around μ.
- For $\sigma^2 = 0$, all the values in data set are equal to zero. The $X \sim N(\mu, \sigma^2)$ is the distribution of constant values μ with probability one at μ.



Normal Distribution

```
>>> import numpy as np
>>> from scipy.stats import norm
>>> import matplotlib.pyplot as plt
>>> x = np.linspace(-7,7,100)
>>> plt.plot(x,norm.pdf(x,loc=0,scale=1),label="mu=0, sigma=1")
>>> plt.plot(x,norm.pdf(x,loc=0,scale=2),label="mu=0, sigma=2")
>>> plt.plot(x,norm.pdf(x,loc=0,scale=2/3),label="mu=0, sigma=2/3")
>>> plt.plot(x,norm.pdf(x,loc=4,scale=2/3),label="mu=4, sigma=2/3")
>>> plt.legend(loc="best")
>>> plt.show()
```



Standard Normal Distribution

The equation of normal curve with μ and σ^2 can be written as

$$y = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}z^2}$$

Where as $z = \frac{x-\mu}{\sigma}$ shows how many standard deviations away the value is from the mean. If the normal distribution has $\mu = 0$ and $\sigma = 1$, we get **standard normal distribution**. The area under standard normal curve can be defined as

$$y = \phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

Which known as **standard normal density function**

Standard Normal Distribution cdf

The standard normal cdf $\Phi(z)$ gives the area under standard normal curve that is left to the value z

$$\Phi(z) = \int_{-\infty}^{z} \phi(y)dy$$

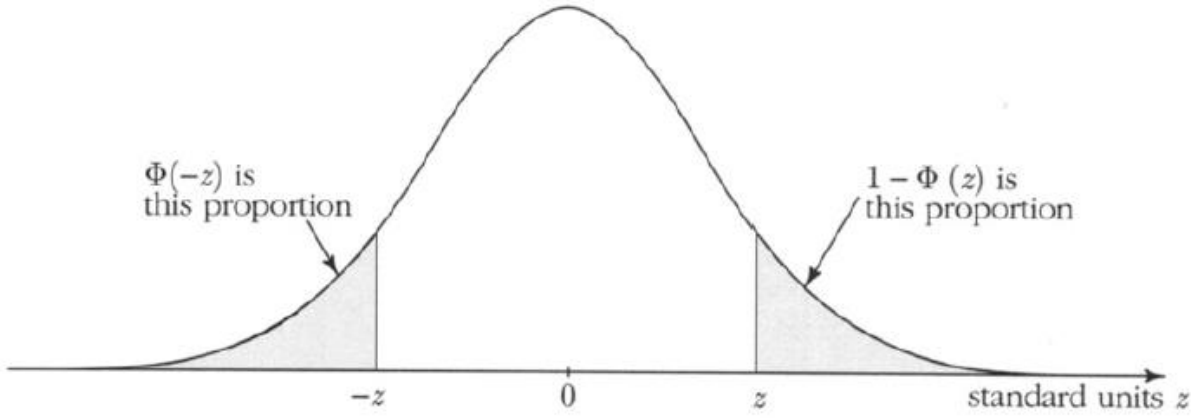
For $Normal(\mu, \sigma^2)$, the probability between a and b is

$$\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

From the symmetry of the normal curve

$$\Phi(-z) = 1 - \Phi(z)$$

For $-\infty < z < \infty$



Standard Normal Distribution

The probability of the interval (a, b) for standard normal distribution can be denoted by

$$\Phi(a, b) = \Phi(b) - \Phi(a)$$

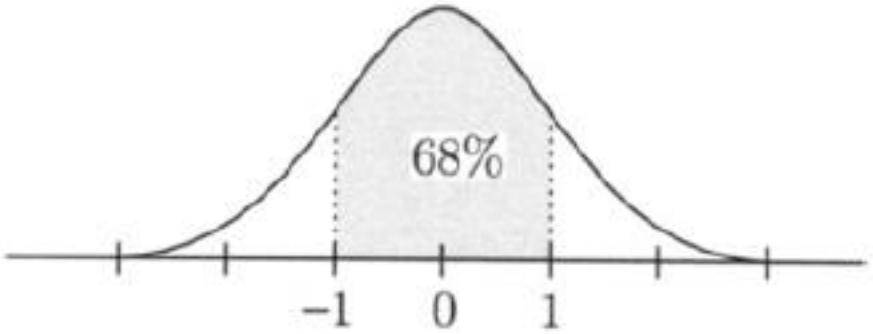
These formulas are used when we work with the normal distribution.

While working with the normal distribution, sketch the standard normal curve and remember the definition of $\Phi(z)$ as the proportion of area under the curve that is left to z.

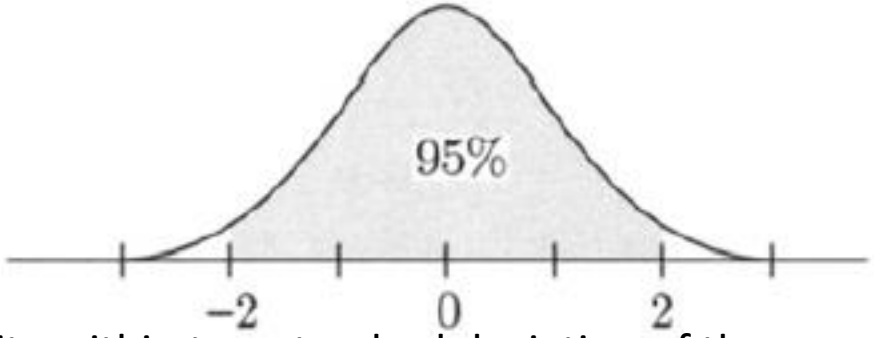
The three most common standard normal probabilities are

Central Limit Theorem

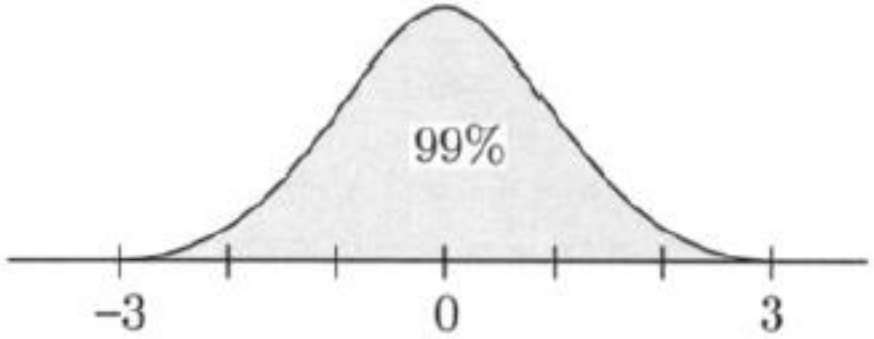
- We can explain the appearance of the normal distribution in several contexts can be explained through central limit theorem.
- For independent random variables having same distribution and finite variance, as number of samples tend to infinity, the distribution of standardized sum (or average) of n variable approaches the standard normal distribution.
- When we average a large number of independent measurements whereas each measurement is smaller than the sum of all the measurements, the distribution of sum approaches the normal shape even if the shape of individual measurements does not follow.



The probability within one standard deviation of the mean $\Phi(-1,1) \approx 68.26\%$



The probability within two standard deviation of the mean $\Phi(-2,2) \approx 95.44\%$



The probability within three standard deviation of the mean $\Phi(-3,3) \approx 99.74\%$

Samatrix.io

Example

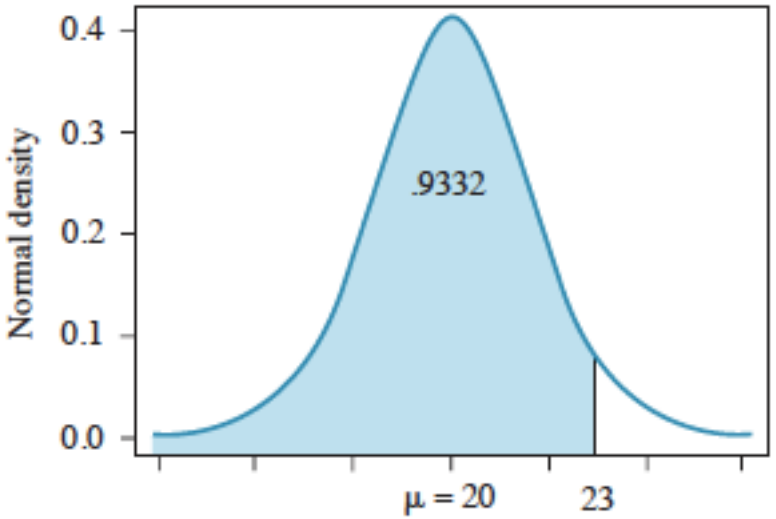
For a normal distribution $X \sim N(20,2^2)$, define the probability that the measurement will be less than 23

First, we calculate the number of standard deviations the value is away from mean

$$z = \frac{x - \mu}{\sigma} = \frac{23 - 20}{2} = 1.5$$

So x lies 1.5 standard deviation away from mean. We can find the area corresponding to $z = 1.5$ as

```
>>> import scipy.stats as st
>>> st.norm.cdf(1.5)
0.9331927987311419
```



The probability that a measurement is less than 23 is 0.933

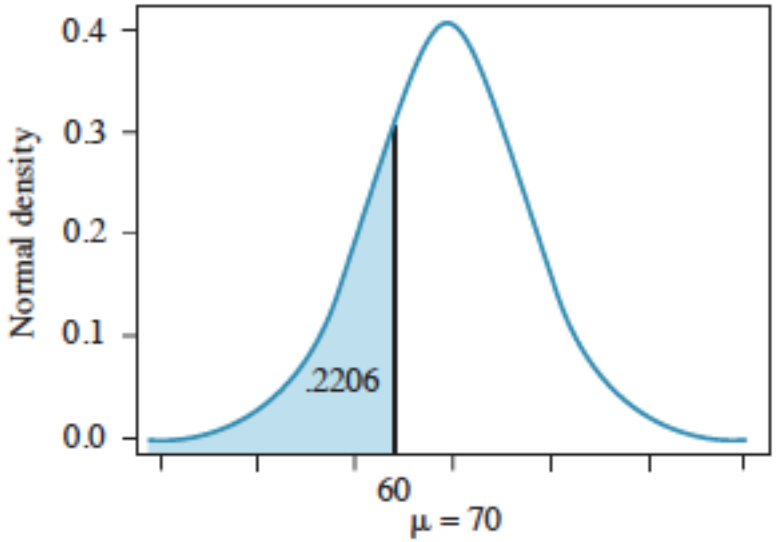
Samatrix.io

Example

For a milk dairy, the mean daily production has a normal distribution $X \sim N(70, 13^2)$.

- What is the probability that the milk production for a randomly chosen cow will be less than 60 pounds?
- What is the probability that the milk production for a randomly chosen cow will be greater than 90 pounds?
- What is the probability that the milk production for a randomly chosen cow will be between 60 and 90 pounds?

Solution (a)



z score is

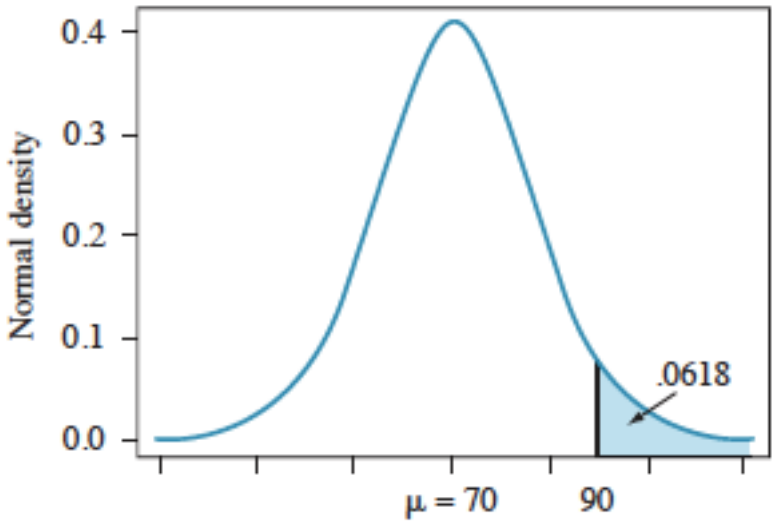
$$z = \frac{60 - 70}{13} = -.77$$

```
>>> import scipy.stats as st
>>> st.norm.cdf(-.77)
0.2206499463
```

Area to the left is 0.2206. So the probability that randomly chosen cow will be less than 60 pounds is 0.2206

Samatrix.io

Solution (b)



z score is

$$z = \frac{90 - 70}{13} = 1.54$$

```
>>> import scipy.stats as st
>>> st.norm.cdf(1.54)
0.9382198232881881
```

Area to the left is 0.9382. So the probability that randomly chosen cow will be more than 90 pounds is $1 - 0.9382 = \ 0.0618$

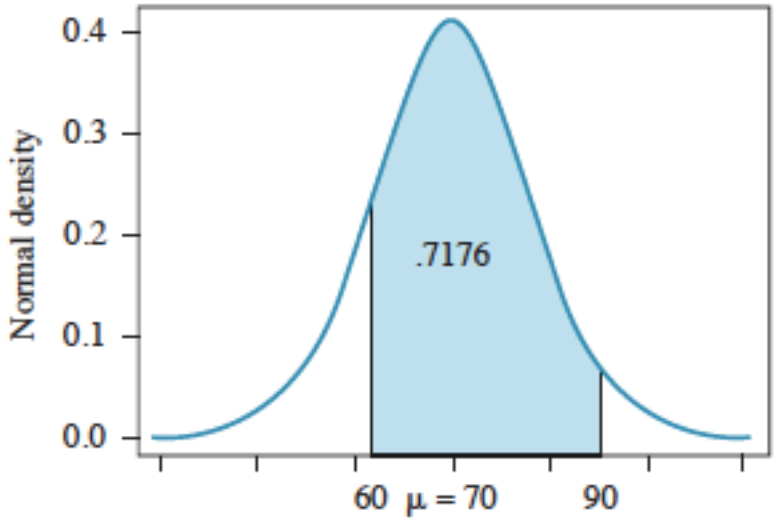
Samatrix.io

Solution (c)

To find the probability between 60 and 90, we have to find area between 60 and 90.

$$0.9382 - 0.2206 = 0.7176$$

So, we can say that the production for 22.06% cow is less than 60 pounds, 71.76% is between 60 and 90 pounds and 6.18% is more than 90 pounds



Samatrix.io

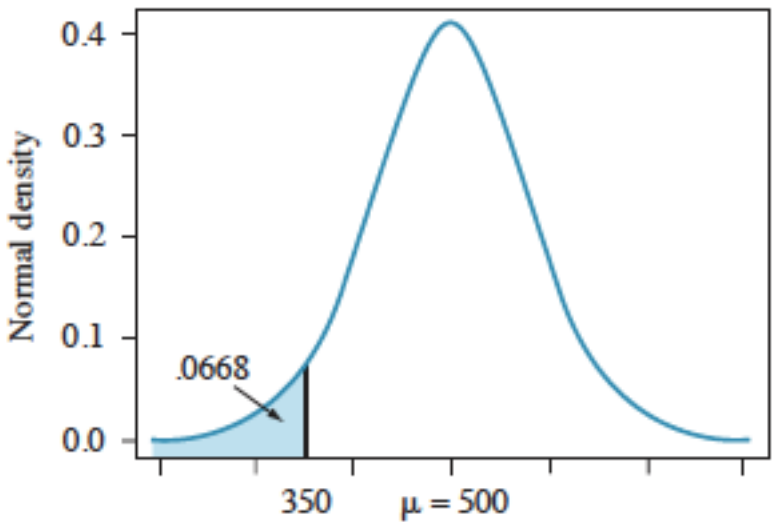
Example

The score for an entrance exam follows the normal distribution with $X \sim N(500, 100^2)$

What proportion of the students taking exam will score below 350. Also calculate lower 10 percentile of all scores

The z value is

$$z = \frac{(350 - 500)}{100} = -1.5$$




```
>>> import scipy.stats as st
>>> st.norm.cdf(-1.5)
0.06680720126885807
```

Samatrix.io

Solution

So, the 6.68% students who scored exam scored less than 350

For second part, we need to find the z value corresponding to probability 10%

```
>>> import scipy.stats as st
>>> st.norm.ppf(.10)
-1.2815515655446004
```

The z score is -1.28

$$z = -1.28 = \frac{x - 500}{100}$$

$$x = (-1.28)100 + 500 = 372$$

So 10 percentile of score is less than 372

Student’s t-distribution

Test Scores are normally distributed with std dev of 5.4,
Sample with 50 random scores were taken with mean of 79 then calculate population test score mean with 95% confidence interval

Samatrix.io

Student's t-distribution

- The Student's t-distribution (or t-distribution) is a member of continuous.
- This distribution is required when the sample size is small and we do not know the population standard deviation.
- William Sealy Gosset under the pseudonym Student developed the t-distribution.
- For sample of n measurements from a normal distribution, the t-distribution with $\nu = n - 1$ degree of freedom is the distribution of location of sample mean relative to true mean, divided by the sample standard deviation and multiplying with $\sqrt{n}$

Z-score and t-value

If $X_1, \ldots, X_n$ are independently and identically drawn from a normally distributed population with mean μ and variance σ^2 i.e. $X \sim N(\mu, \sigma^2)$.

The sample mean would be

$$\bar{X} = \frac{1}{n}\sum_{i=i}^{n} X_i$$

The sample variance is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

When the population standard deviation is known, the z-score is given by

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

With sample standard deviation and $n - 1$ degree of freedom, the t-score is given by

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

The degree of freedom

- For a dynamic system, the number of degrees of freedom is the minimum number of independent coordinated that are used to completely specify the position of the system.
- For statistics, the number of degrees of freedom is the number of measurements that are free to vary in the calculation of statistic.
- The degrees of freedom can also be defined as the number of independent measurements of a sample of data that we can use to estimate a parameter of population from which the sample is drawn.
- For example, if we have n measurements, for mean, we have n independent observation and degrees of freedom is n. For variance, one degrees of freedom is lost to calculate μ, so we have $n - 1$ degrees of freedom.

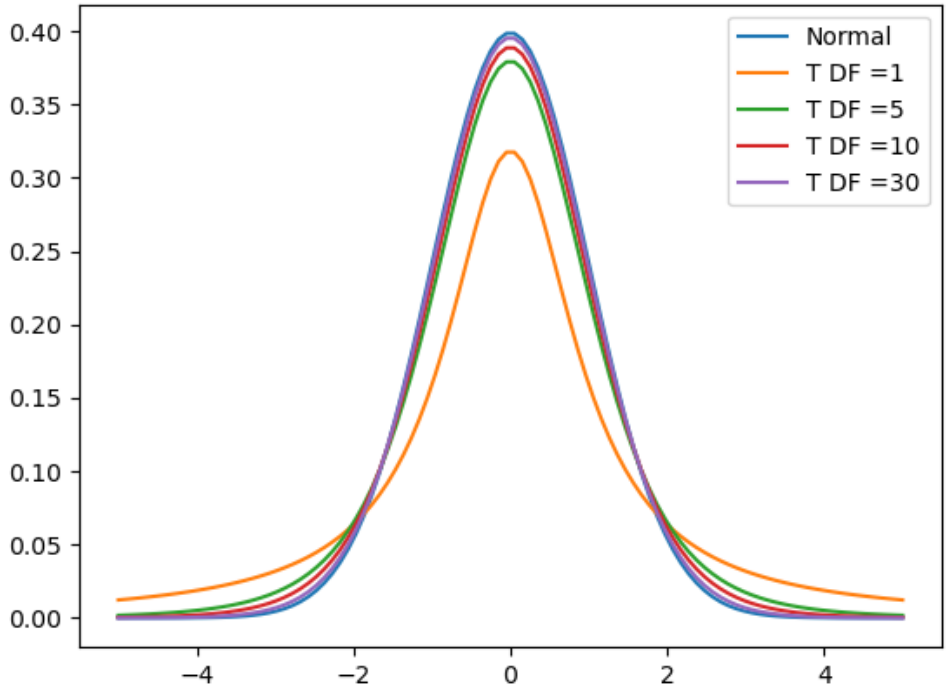
Normal distribution vs t-distribution

- Initially the statistics focused on finding probability and inference using large samples and the normal distribution. The standard normal distribution results into the bell-shaped probability distribution for large samples but for small samples results into larger probability areas in the trails of the distribution.
- The standard normal distribution and student t-distribution both are symmetrical and have a mean of zero. The standard normal distribution is bell shape and has a standard deviation of one whereas t-distribution is uni-model and has a standard deviation that is not equal to one.
- The standard deviation of the t-distribution varies. For a small sample size, the t-distribution is more peaked (leptokurtic). Compared to the standard normal distribution, t-distribution's probability areas in the tail are higher. In other words the probability density of the t-distribution is lower in the centre and heavier in the tail.

Normal distribution vs t-distribution

Following graph shows the density of t-distribution with an increase in degrees of freedom ν. With the increase in the value of ν, the t-distribution becomes closer to normal distribution

```
>>> import numpy as np
>>> from scipy.stats import norm
>>> from scipy.stats import t
>>> import matplotlib.pyplot as plt
>>> x = np.linspace(-5,5,100)
>>> plt.plot(x,norm.pdf(x),label="Normal")
>>> plt.plot(x,t.pdf(x,1), label="T DF =1")
>>> plt.plot(x,t.pdf(x,5), label="T DF =5")
>>> plt.plot(x,t.pdf(x,10), label="T DF =10")
>>> plt.plot(x,t.pdf(x,30), label="T DF =30")
>>> plt.legend(loc="best")
>>> plt.show()
```



Samatrix.io

X = 82

Mu = 6.5

n = 100

90% Confidence interval

Samatrix.io

Exponential Distribution

- Let’s study a continuous distribution function which is closely related to Poisson distribution.
- For Poisson point process, we count the number of occurrences in a given interval.
- This is a discrete type random variable and follows Poisson distribution.
- Along with the number of occurrences, waiting time (or distance) between successive occurrences is also a random variable.
- For example, waiting time between the arrival of email; or the waiting time between phone calls at a telephone exchange; or the locations of road accidents on a national highway.

Exponential Distribution

- The distance or span of time between two consecutive point is a continuous random variable X.
- The random variable can take any positive value.
- It forms the **exponential distribution**.
- For the exponential distribution there is only one parameter $\lambda > 0$
- The exponential distribution is often used to model the failure of the objects.
- The failures form a Poisson process in time and the time to next failure is exponentially distributed.

Exponential Distribution

Exponential Distribution: The probability density function of a continuous random variable X that follows exponential distribution is

$$P(X|\lambda) = \begin{cases} \lambda e^{-\lambda x} & for\ x \geq 0 \\ 0 & otherwise \end{cases}$$

Where $\lambda > 0$ is a parameter

Mean and Variance of the Exponential Distribution

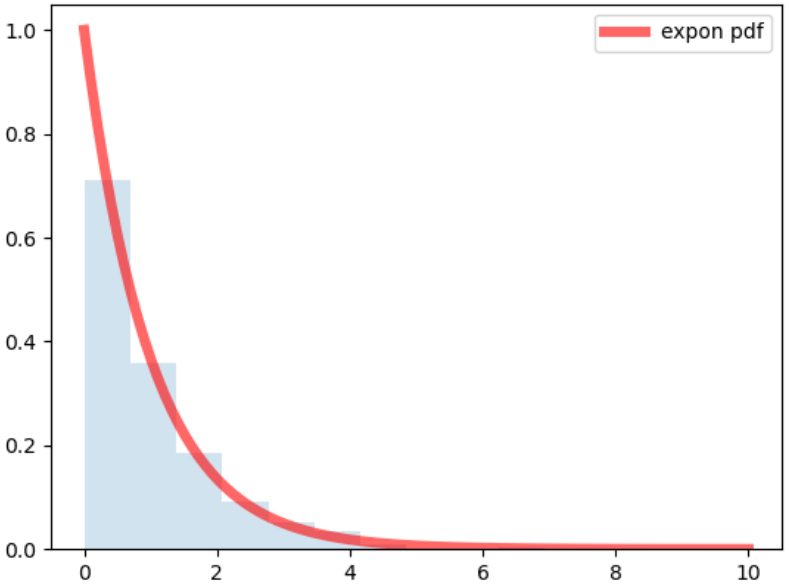
An exponential distribution with parameter λ has

Mean: - $1/\lambda$

Variance: - $1/\lambda^2$

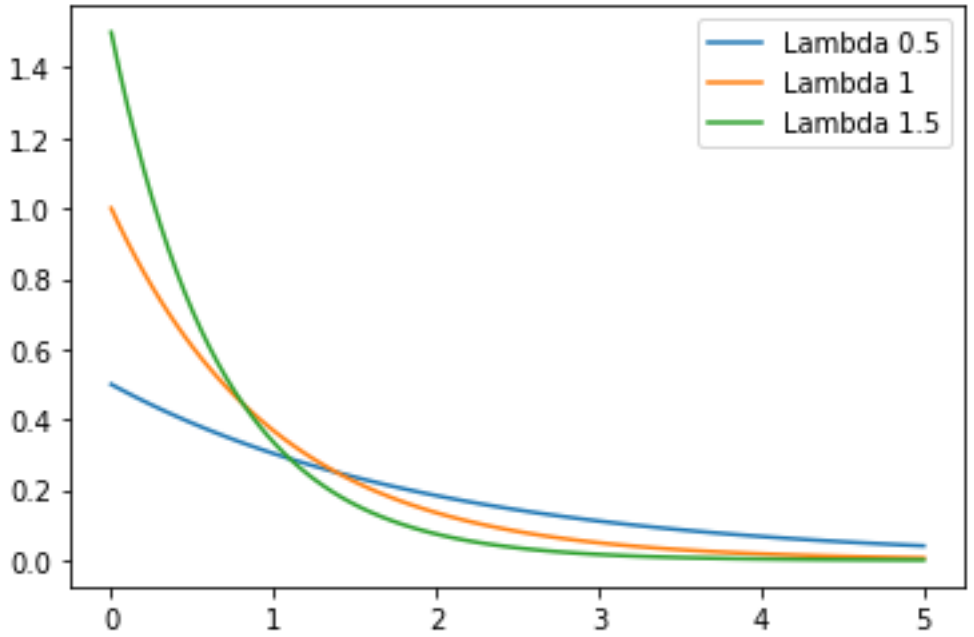
Exponential Distribution using Python

```
>>> from scipy.stats import expon
>>> import matplotlib.pyplot as plt
>>> fig, ax = plt.subplots(1, 1)
>>> from scipy.stats import expon
>>> import matplotlib.pyplot as plt
>>> import numpy as np
>>> fig, ax = plt.subplots(1, 1)
>>> x = np.linspace(0,10,100)
>>> ax.plot(x,expon.pdf(x),'r-',lw=5, alpha=0.6, label='expon pdf')
>>> r=expon.rvs(size=1000)
>>> ax.hist(r, density=True, histtype='stepfilled',alpha=0.2)
>>> ax.legend()
>>> plt.show()
```



Exponential PDF for different lambda

```
from scipy.stats import expon
import matplotlib.pyplot as plt
import numpy as np
fig, ax = plt.subplots(1, 1)
x = np.linspace(0,5,100)
lambda1, lambda2, lambda3 = 1/0.5,1,1/1.5
ax.plot(x,expon.pdf(x,scale=lambda1), label='Lambda 0.5')
ax.plot(x,expon.pdf(x,scale=lambda2), label='Lambda 1')
ax.plot(x,expon.pdf(x,scale=lambda3), label='Lambda 1.5')
ax.legend()
plt.show()
```



- In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean of 25 log-ons per hour. What is the probability that there are no log-ons in an interval of six minutes?
- What is the probability that the time until the next log-on is between two and three minutes?

Beta family of distribution

The $beta(a, b)$ distribution is another important distribution that is used for continuous random variable x in the range of $0 \leq x \leq 1$. The beta distribution has the probability function of the form

$$P(x|\alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1 - x)^{\beta-1}$$

Where $\alpha > 0$ and $\beta > 0$

The shape of the curve that is determined by $x^{\alpha-1}(1 - x)^{\beta-1}$ is called $beta(\alpha, \beta)$. The constant $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$ is required to make the curve a probability density function.

The curve can have different shapes depending on the values a and b. Which makes the $beta(a, b)$ a family of distributions. The $uniform(0,1)$ is a special case of $beta(a, b)$ where $a = 1$ and $b = 1$

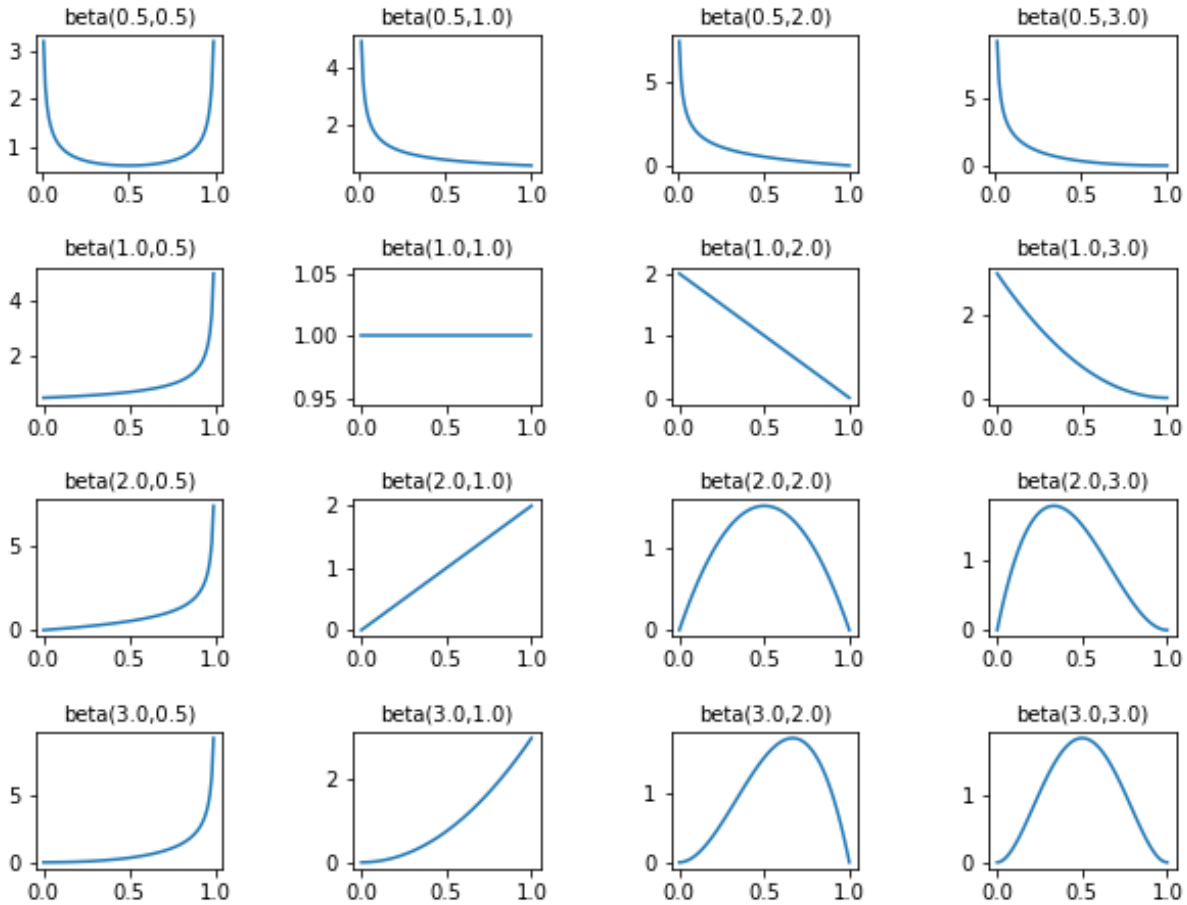
Biased coin flipped 50 times

Heads 30 times, tails 20 times

Plot beta distribution

Beta family of distribution

- The figure shows the beta distribution $beta(a, b)$ for the value's different values of a, b that the beta family can take.
- For $a < b$, the density has more weight in the lower half.
- When the $a > b$, the opposite is true.
- For $a = b$, the $beta(a, b)$ density is symmetric.
- For $a = 1/2$, more weight is given towards zero and for $b = 1/2$, more weight is given towards one.
- For $a = 1, b = 1$, you can see the uniform distribution.



Samatrix.io

Beta Distribution using Python

```
import numpy as np
from scipy.stats import beta
import matplotlib.pyplot as plt
fig, ((ax1, ax2, ax3, ax4), (ax5, ax6, ax7, ax8), (ax9, ax10, ax11, ax12),(ax13, ax14, ax15,
ax16)) = plt.subplots(4, 4)
fig.set_size_inches(10,7.5)
x = np.linspace(0,1,100)
ax1.plot(x,beta.pdf(x, 0.5, 0.5))
ax2.plot(x,beta.pdf(x, 0.5, 1.0))
ax3.plot(x,beta.pdf(x, 0.5, 2.0))
ax4.plot(x,beta.pdf(x, 0.5, 3.0))
ax5.plot(x,beta.pdf(x, 1.0, 0.5))
```

Beta Distribution using Python

```
ax6.plot(x,beta.pdf(x, 1.0, 1.0))
ax7.plot(x,beta.pdf(x, 1.0, 2.0))
ax8.plot(x,beta.pdf(x, 1.0, 3.0))
ax9.plot(x,beta.pdf(x, 2.0, 0.5))
ax10.plot(x,beta.pdf(x, 2.0, 1.0))
ax11.plot(x,beta.pdf(x, 2.0, 2.0))
ax12.plot(x,beta.pdf(x, 2.0, 3.0))
ax13.plot(x,beta.pdf(x, 3.0, 0.5))
ax14.plot(x,beta.pdf(x, 3.0, 1.0))
ax15.plot(x,beta.pdf(x, 3.0, 2.0))
ax16.plot(x,beta.pdf(x, 3.0, 3.0))
```

Beta Distribution using Python

```
ax1.set_title("beta(0.5,0.5)",fontsize=10)
ax2.set_title("beta(0.5,1.0)",fontsize=10)
ax3.set_title("beta(0.5,2.0)",fontsize=10)
ax4.set_title("beta(0.5,3.0)",fontsize=10)
ax5.set_title("beta(1.0,0.5)",fontsize=10)
ax6.set_title("beta(1.0,1.0)",fontsize=10)
ax7.set_title("beta(1.0,2.0)",fontsize=10)
ax8.set_title("beta(1.0,3.0)",fontsize=10)
ax9.set_title("beta(2.0,0.5)",fontsize=10)
ax10.set_title("beta(2.0,1.0)",fontsize=10)
ax11.set_title("beta(2.0,2.0)",fontsize=10)
ax12.set_title("beta(2.0,3.0)",fontsize=10)
ax13.set_title("beta(3.0,0.5)",fontsize=10)
ax14.set_title("beta(3.0,1.0)",fontsize=10)
ax15.set_title("beta(3.0,2.0)",fontsize=10)
ax16.set_title("beta(3.0,3.0)",fontsize=10)
fig.subplots_adjust(hspace=0.7)
fig.subplots_adjust(wspace=0.7)
plt.show()
```

Beta Distribution Mean and Variance

A Beta Distribution with parameters a, b

Mean $\frac{a}{a+b}$

Variance $\frac{ab}{(a+b)^2(a+b+1)}$

Gamma family of distributions

The $gamma(a, b)$ distribution is used for a non-negative continuous random variable $0 \leq x < \infty$. The two parameters $a > 0, b > 0$

$$P_\gamma(x|a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$$

The shape of the curve is given by $x^{a-1}e^{-bx}$. The constant $\frac{b^a}{\Gamma(a)}$ is required to make the curve a probability density.

In scipy, the parameter a, b are given as

y1 = stats.gamma.pdf(x, a, scale=1/b)

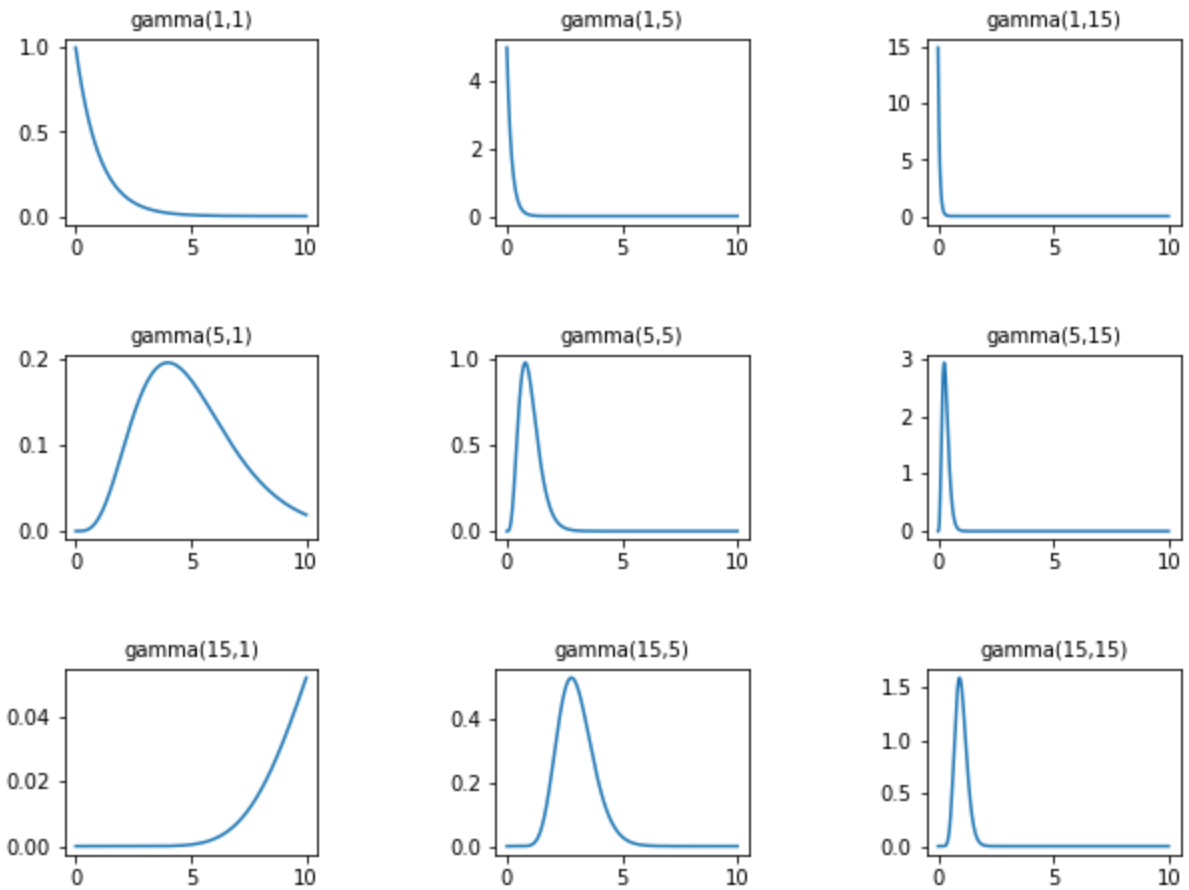
Gamma family of distributions - Python

```
import numpy as np
from scipy.stats import gamma
import matplotlib.pyplot as plt
fig, ((ax1, ax2, ax3), (ax4, ax5, ax6), (ax7, ax8, ax9)) = plt.subplots(3, 3)
fig.set_size_inches(10,7.5)
x = np.linspace(0,10,1000)
ax1.plot(x,gamma.pdf(x, 1.0, scale=1/1))
ax2.plot(x,gamma.pdf(x, 1.0, scale=1/5))
ax3.plot(x,gamma.pdf(x, 1.0, scale=1/15))
ax4.plot(x,gamma.pdf(x, 5.0, scale=1/1))
ax5.plot(x,gamma.pdf(x, 5.0, scale=1/5))
ax6.plot(x,gamma.pdf(x, 5.0, scale=1/15))
ax7.plot(x,gamma.pdf(x, 15.0, scale=1/1))
ax8.plot(x,gamma.pdf(x, 15.0, scale=1/5))
ax9.plot(x,gamma.pdf(x, 15.0, scale=1/15))
```

Gamma family of distributions - Python

```
ax1.set_title("gamma(1,1)",fontsize=10)
ax2.set_title("gamma(1,5)",fontsize=10)
ax3.set_title("gamma(1,15)",fontsize=10)
ax4.set_title("gamma(5,1)",fontsize=10)
ax5.set_title("gamma(5,5)",fontsize=10)
ax6.set_title("gamma(5,15)",fontsize=10)
ax7.set_title("gamma(15,1)",fontsize=10)
ax8.set_title("gamma(15,5)",fontsize=10)
ax9.set_title("gamma(15,15)",fontsize=10)

fig.subplots_adjust(hspace=0.7)
fig.subplots_adjust(wspace=0.7)
plt.show()
```



Gamma family of distributions

Mean and Variance of $gamma(a, b)$

Mean - a/b

Variance - a/b^2

Chi-Square distributions

The chi-square $\chi^2(r)$ distribution with r degree of freedom is a special case of $gamma$ distribution with $a = \frac{r}{2}, b = \frac{1}{2}$

$$P(x|r) = \frac{1}{\Gamma(\mathrm{r}\backslash 2)2^{\frac{\mathrm{r}}{2}}} x^{\frac{r}{2}-1} e^{-\frac{x}{2}}, \qquad 0 < x < \infty$$

Mean and Variance for Chi-Square Distribution

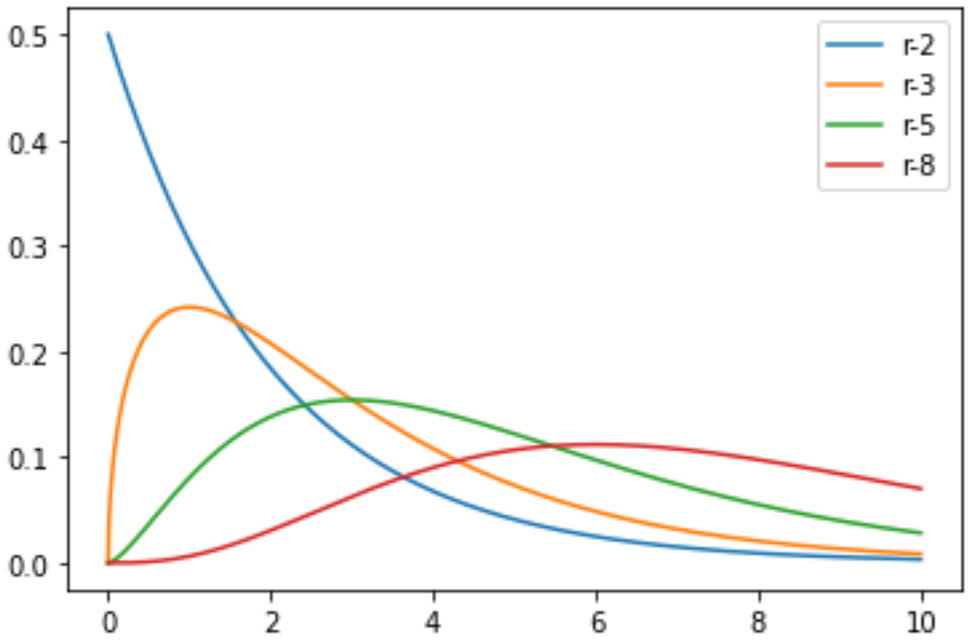
Mean r

Variance $2r$

Samatrix.io

Chi-Square distributions

```
import numpy as np
from scipy.stats import chi2
import matplotlib.pyplot as plt
fig, ax = plt.subplots(1, 1)
x = np.linspace(0,10,1000)
ax.plot(x,chi2.pdf(x, 2),label="r-2")
ax.plot(x,chi2.pdf(x, 3),label="r-3")
ax.plot(x,chi2.pdf(x, 5),label="r-5")
ax.plot(x,chi2.pdf(x, 8),label="r-8")
ax.legend()
plt.show()
```



Joint Probability Distribution

- The joint probability distribution of two continuous random variables X and Y can be specified by providing a method for calculating the probability that X and Y assume a value in any region R of two-dimensional space.
- The joint probability density function is defined over two-dimensional space.
- The probability that (X, Y) assumes a value in region R can be calculated using double integral of $f_{XY}(x, y)$ over a region R.
- The integral can be interpreted as the volume under the surface over the region R

Joint Probability Distribution

The joint probability density function for the continuous random variable X and Y denoted as $f_{XY}(x, y)$ satisfy the following property

1. $f_{XY}(x, y) \geq 0$ for all x, y
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx\, dy = 1$
3. For any region R of two-dimensional space

$$P([X, Y] \in R = \int_R \int f_{XY}(x, y)\, dx\, dy$$

Quantile-Quantile Plot

- We use Quantile-Quantile or Q-Q plot to compare two probability distributions by plotting their quantiles against each other.
- This is a technique to check whether two sets of sample points follow the same distribution.
- We use the qqplots to check whether the sample of data follow a particular distribution.
- In this scenario, one distribution is known and other distribution is unknown.
- If the other distribution follows the given distribution, we will have a scatter-plot where the data points will be in a straight line.
- If the distributions are identical, the quantities should be approximately equal.

Quantile-Quantile Plot

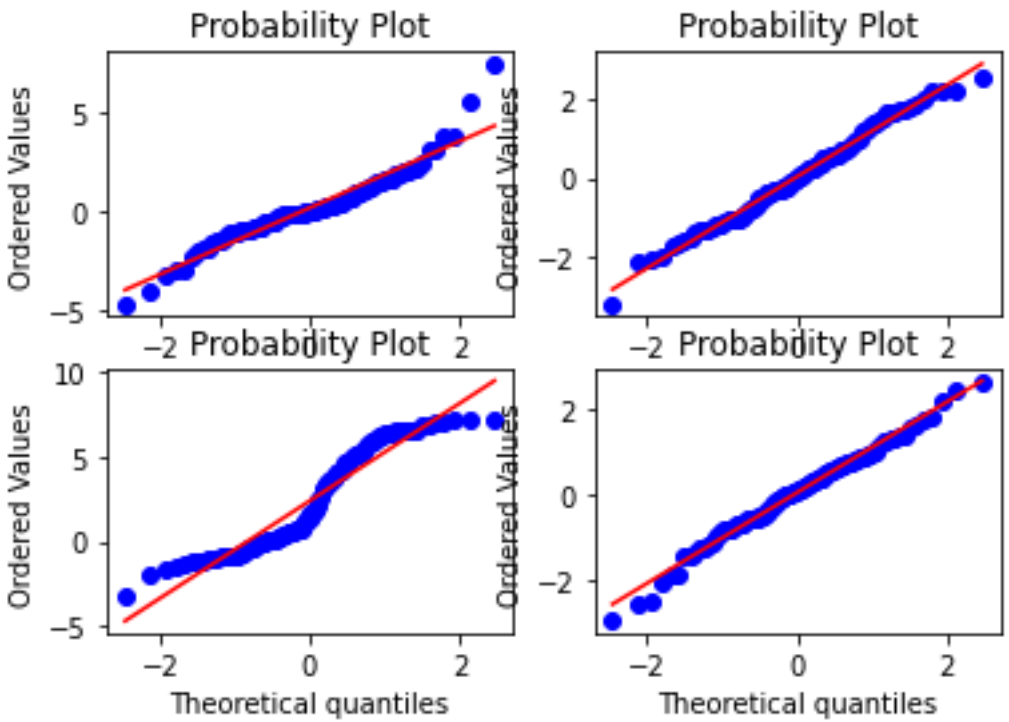
```
from scipy import stats
import matplotlib.pyplot as plt
nsample = 100
np.random.seed(7654321)

# t-plot for small degree of freedom
ax1 = plt.subplot(221)
x = stats.t.rvs(3, size=nsample)
res = stats.probplot(x, plot=plt)

# t-plot for large degree of freedom
ax2 = plt.subplot(222)
x = stats.t.rvs(25, size=nsample)
res = stats.probplot(x, plot=plt)
```

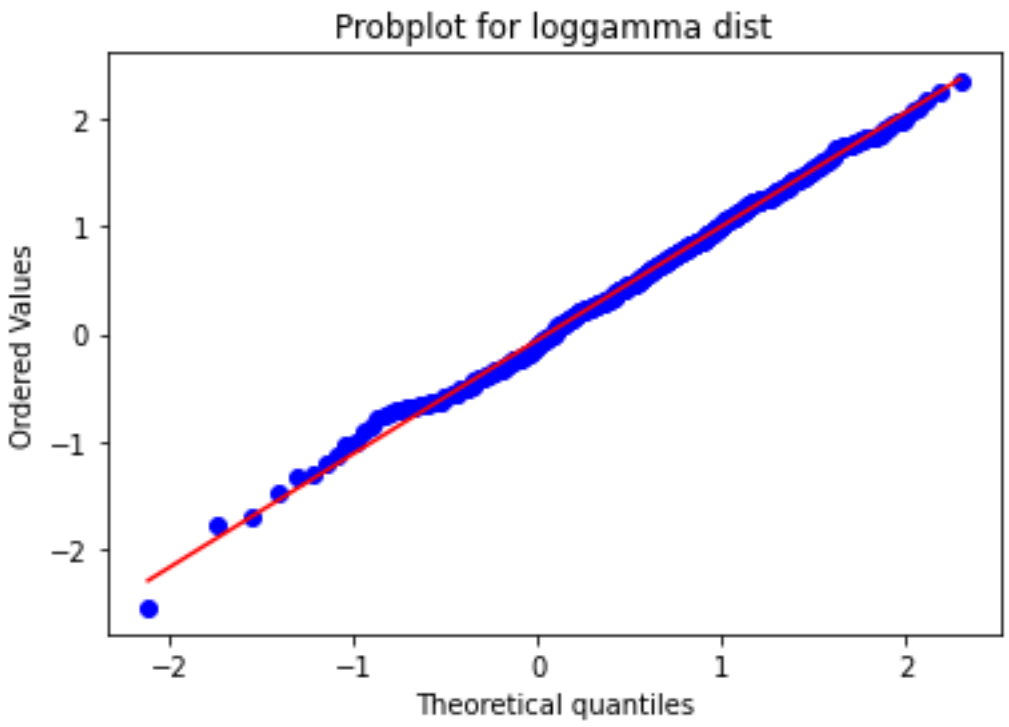
Quantile-Quantile Plot

```
#Two normal distribution with broadcasting
ax3 = plt.subplot(223)
x = stats.norm.rvs(loc=[0,5],
scale=[1,1.5],size=(nsample//2,2)).ravel()
res = stats.probplot(x, plot=plt)
# Standard normal distribution
ax4 = plt.subplot(224)
x = stats.norm.rvs(loc=0, scale=1,
size=nsample)
res = stats.probplot(x, plot=plt)
plt.show()
```



Quantile-Quantile Plot

```
#Loggamma distribution
fig = plt.figure()
ax = fig.add_subplot(111)
x = stats.loggamma.rvs(c=2.5, size=500)
res = stats.probplot(x, dist=stats.loggamma, sparams=(2.5,), plot=ax)
ax.set_title("Probplot for loggamma dist")
plt.show()
```



Samatrix.io

Thanks

Samatrix Consulting Pvt Ltd

Samatrix.io